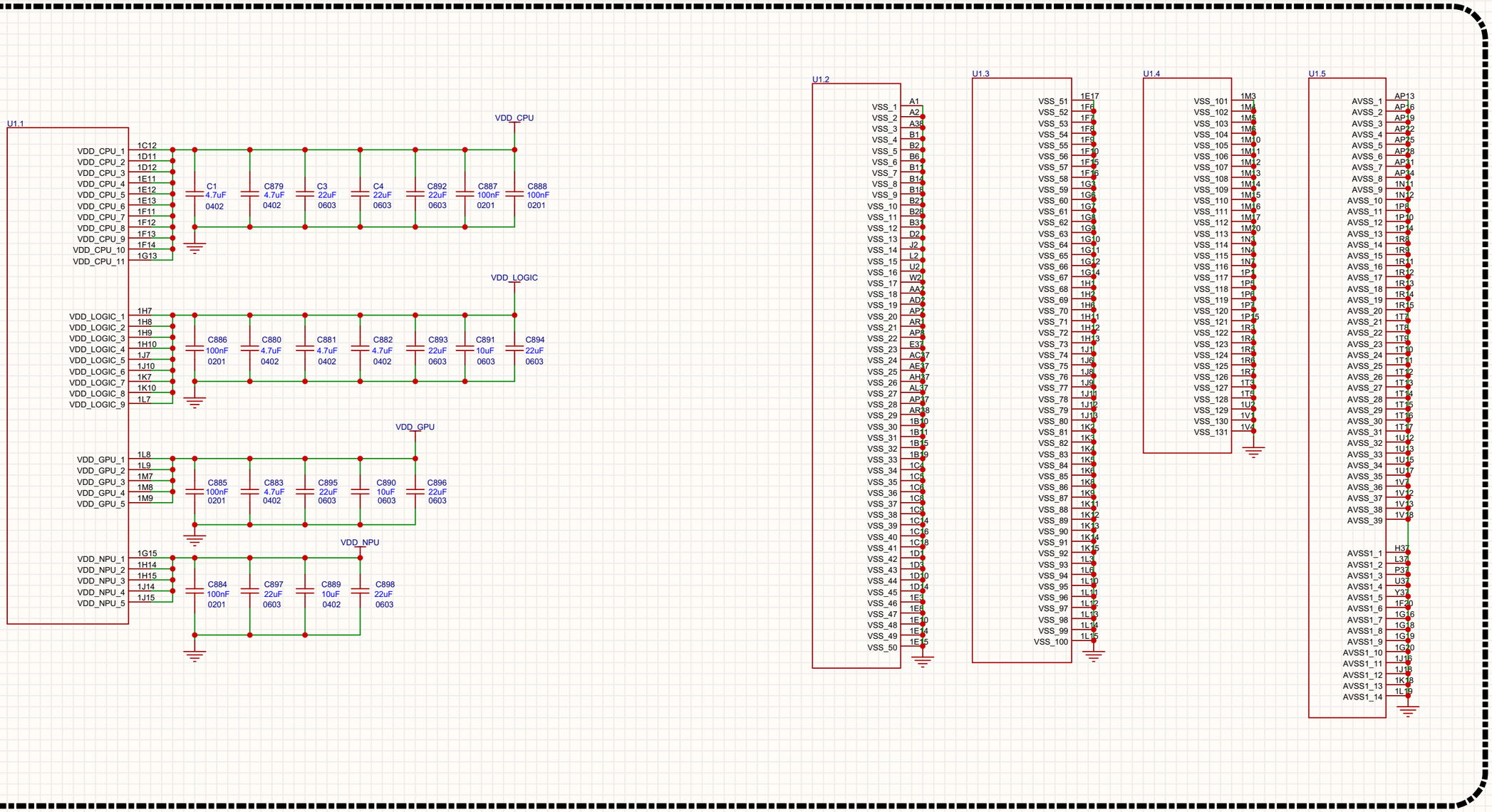
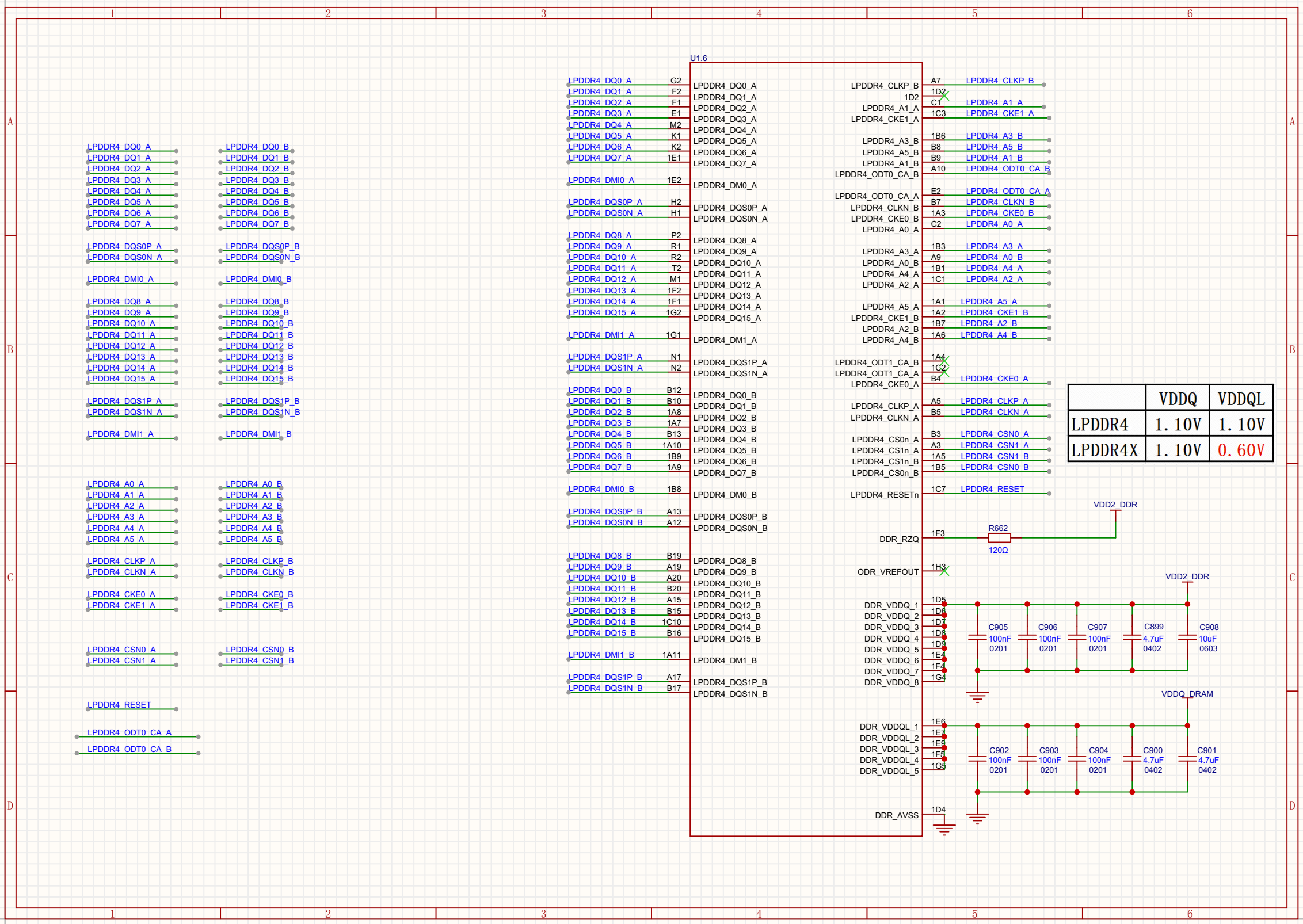
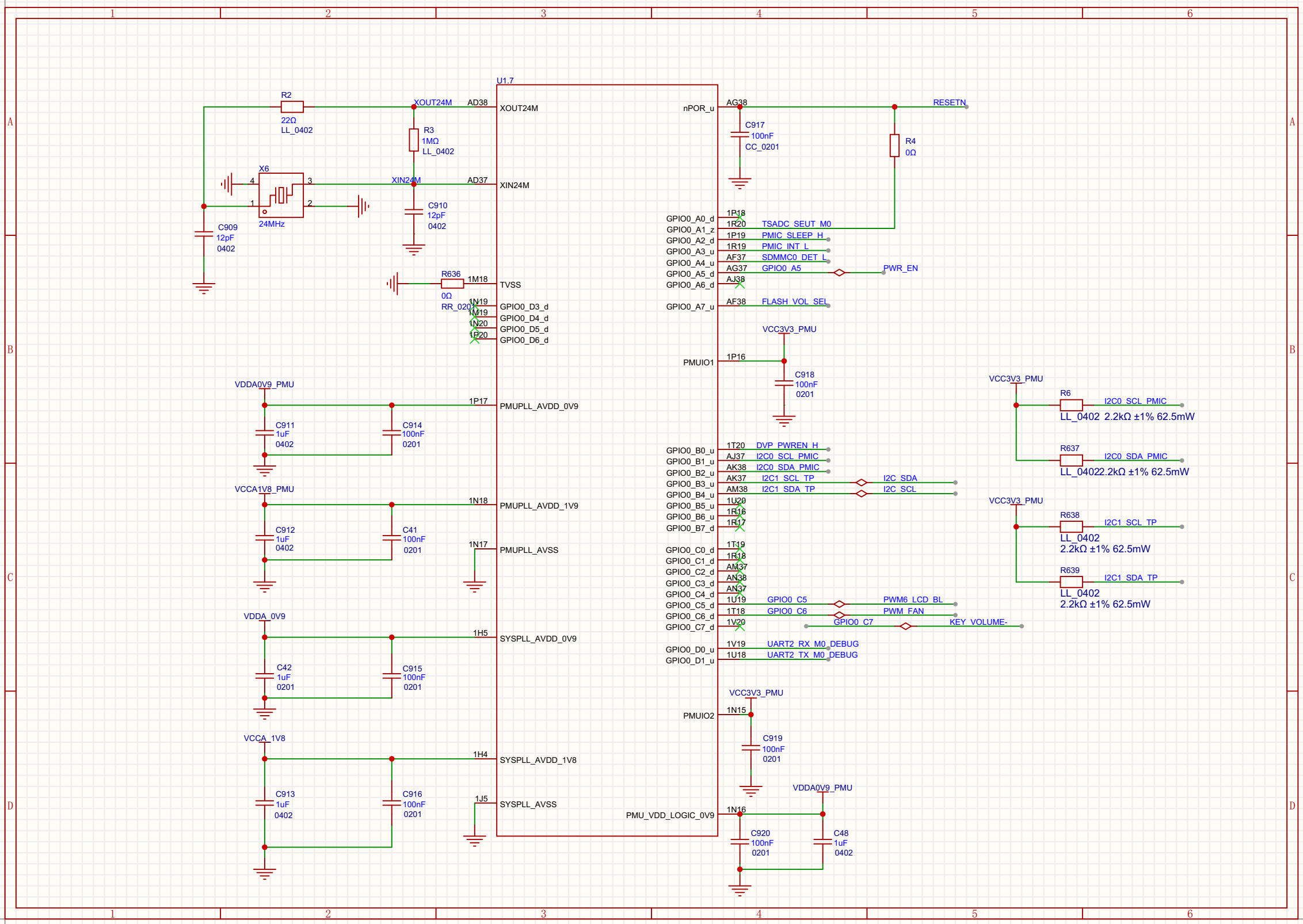
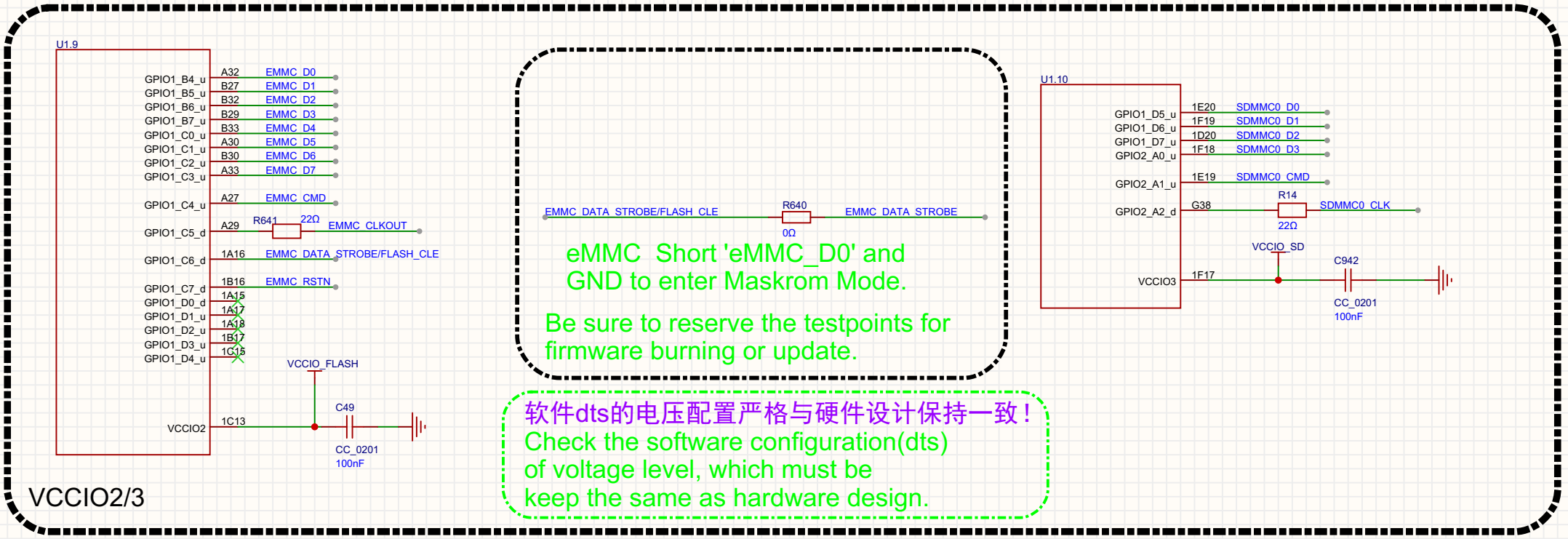
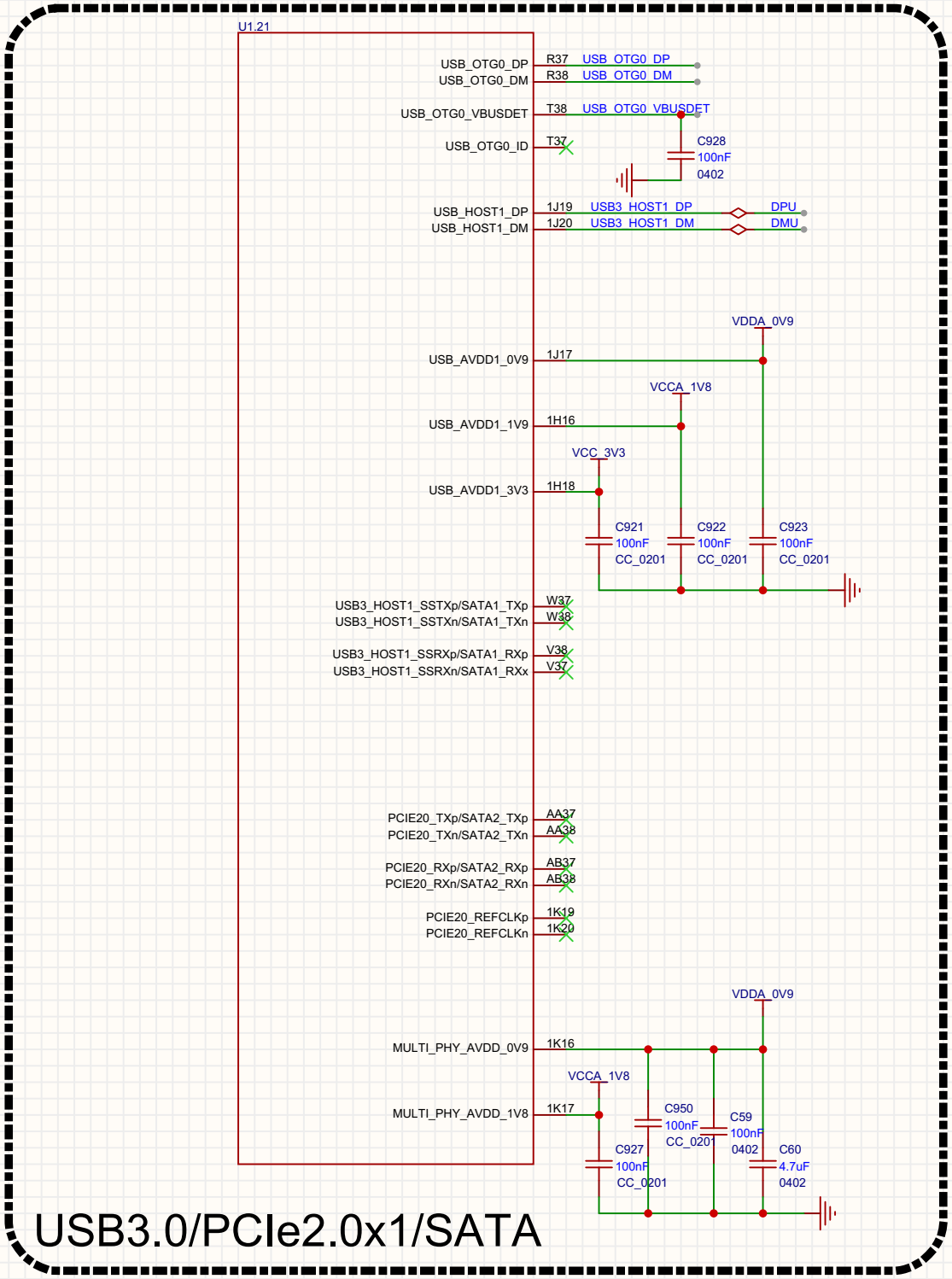
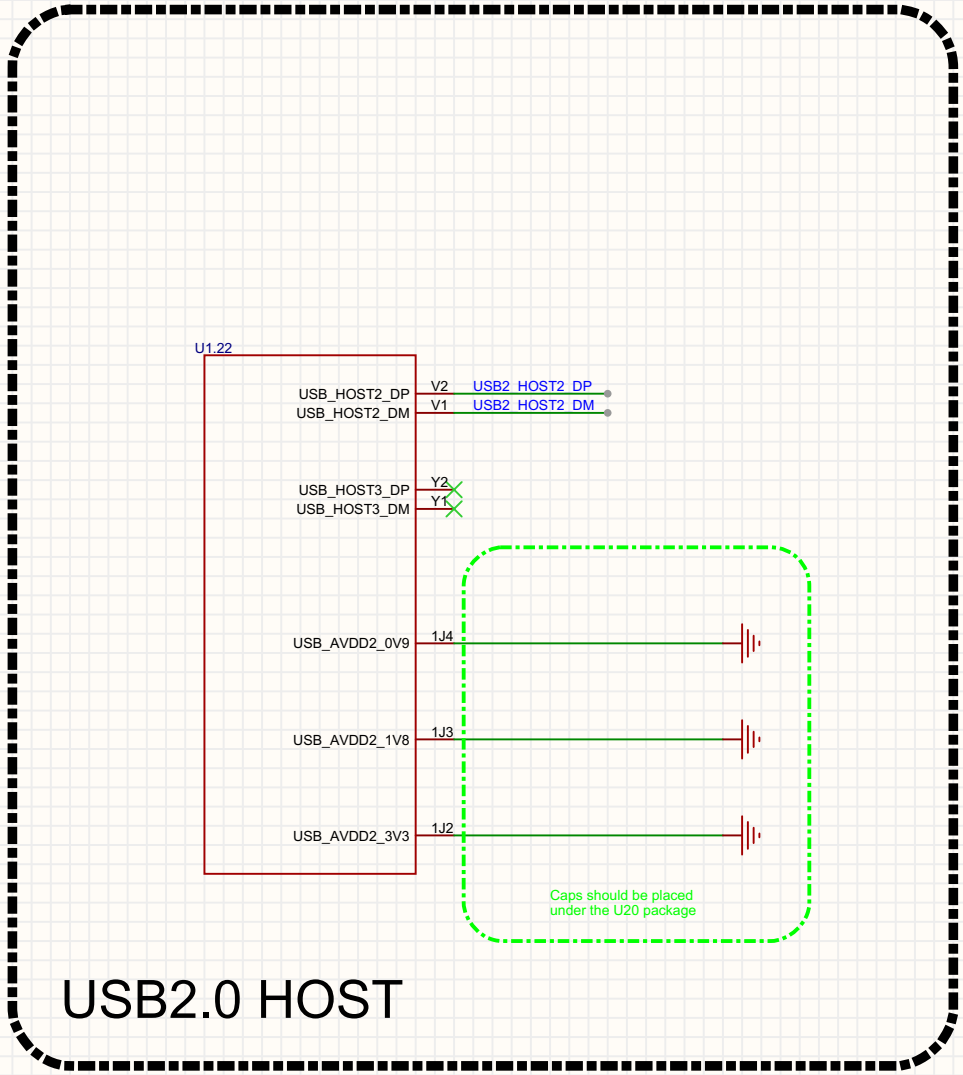

U1.1
VDD_CPU_1
VDD_CPU_2
VDD_CPU_3
VDD_CPU_4
VDD_CPU_5
VDD_CPU_6
VDD_CPU_7
VDD_CPU_8
VDD_CPU_9
VDD_CPU_10
VDD_CPU_11
1C12
1D11
1D12
1E11
1E12
1E13
1F11
1F12
1F13
1F14
1G13
VDD_CPU
C1 4.7uF 0402
C879 4.7uF 0402
C3 22uF 0603
C4 22uF 0603
C892 22uF 0603
C887 100nF 0201
C888 100nF 0201
VDD_LOGIC_1
VDD_LOGIC_2
VDD_LOGIC_3
VDD_LOGIC_4
VDD_LOGIC_5
VDD_LOGIC_6
VDD_LOGIC_7
VDD_LOGIC_8
VDD_LOGIC_9
1H7
1H8
1H9
1H10
1J7
1J10
1K7
1K10
1L7
VDD_LOGIC
C886 100nF 0201
C880 4.7uF 0402
C881 4.7uF 0402
C882 4.7uF 0402
C893 22uF 0603
C891 10uF 0603
C894 22uF 0603
VDD_GPU_1
VDD_GPU_2
VDD_GPU_3
VDD_GPU_4
VDD_GPU_5
1L8
1L9
1M7
1M8
1M9
VDD_GPU
C885 100nF 0201
C883 4.7uF 0402
C895 22uF 0603
C890 10uF 0603
C896 22uF 0603
VDD_NPU_1
VDD_NPU_2
VDD_NPU_3
VDD_NPU_4
VDD_NPU_5
1G15
1H14
1H15
1J14
1J15
VDD_NPU
C884 100nF 0201
C897 22uF 0603
C889 10uF 0402
C898 22uF 0603
U1.2
VSS_1 A1
VSS_2 A2
VSS_3 A38
VSS_4 B1
VSS_5 B2
VSS_6 B6
VSS_7 B11
VSS_8 B14
VSS_9 B18
VSS_10 B21
VSS_11 B25
VSS_12 B31
VSS_13 D2
VSS_14 J2
VSS_15 L2
VSS_16 U2
VSS_17 W2
VSS_18 AA2
VSS_19 AD2
VSS_20 AP2
VSS_21 AR2
VSS_22 AP3
VSS_23 E37
VSS_24 AC37
VSS_25 AE37
VSS_26 AH37
VSS_27 AL37
VSS_28 AP37
VSS_29 AR38
VSS_30 1B10
VSS_31 1B11
VSS_32 1B15
VSS_33 1B19
VSS_34 1C4
VSS_35 1C8
VSS_36 1C9
VSS_37 1C13
VSS_38 1C14
VSS_39 1C16
VSS_40 1C18
VSS_41 1D1
VSS_42 1D5
VSS_43 1D10
VSS_44 1D14
VSS_45 1E3
VSS_46 1E8
VSS_47 1E10
VSS_48 1E14
VSS_49 1E15
VSS_50 1E17
U1.3
VSS_51 1E17
VSS_52 1F6
VSS_53 1F7
VSS_54 1F8
VSS_55 1F9
VSS_56 1F10
VSS_57 1F15
VSS_58 1F16
VSS_59 1G4
VSS_60 1G6
VSS_61 1G7
VSS_62 1G8
VSS_63 1G9
VSS_64 1G10
VSS_65 1G11
VSS_66 1G12
VSS_67 1G14
VSS_68 1H1
VSS_69 1H2
VSS_70 1H6
VSS_71 1H11
VSS_72 1H12
VSS_73 1H13
VSS_74 1J1
VSS_75 1J6
VSS_76 1J8
VSS_77 1J9
VSS_78 1J11
VSS_79 1J12
VSS_80 1J16
VSS_81 1K2
VSS_82 1K3
VSS_83 1K4
VSS_84 1K5
VSS_85 1K6
VSS_86 1K8
VSS_87 1K9
VSS_88 1K11
VSS_89 1K12
VSS_90 1K13
VSS_91 1K14
VSS_92 1K15
VSS_93 1L3
VSS_94 1L6
VSS_95 1L10
VSS_96 1L11
VSS_97 1L12
VSS_98 1L13
VSS_99 1L14
VSS_100 1L15
U1.4
VSS_101 1M3
VSS_102 1M4
VSS_103 1M5
VSS_104 1M6
VSS_105 1M10
VSS_106 1M11
VSS_107 1M12
VSS_108 1M13
VSS_109 1M14
VSS_110 1M15
VSS_111 1M16
VSS_112 1M17
VSS_113 1M20
VSS_114 1N3
VSS_115 1N4
VSS_116 1N7
VSS_117 1P1
VSS_118 1P5
VSS_119 1P6
VSS_120 1P7
VSS_121 1P15
VSS_122 1R4
VSS_123 1R5
VSS_124 1R6
VSS_125 1R7
VSS_126 1R8
VSS_127 1T3
VSS_128 1T5
VSS_129 1U2
VSS_130 1V1
VSS_131 1V4
U1.5
AVSS_1 AP13
AVSS_2 AP16
AVSS_3 AP19
AVSS_4 AP22
AVSS_5 AP25
AVSS_6 AP28
AVSS_7 AP31
AVSS_8 AP34
AVSS_9 1N11
AVSS_10 1N12
AVSS_11 1P8
AVSS_12 1P10
AVSS_13 1P14
AVSS_14 1R8
AVSS_15 1R9
AVSS_16 1R11
AVSS_17 1R12
AVSS_18 1R13
AVSS_19 1R14
AVSS_20 1R15
AVSS_21 1T4
AVSS_22 1T8
AVSS_23 1T9
AVSS_24 1T10
AVSS_25 1T11
AVSS_26 1T12
AVSS_27 1T13
AVSS_28 1T14
AVSS_29 1T15
AVSS_30 1T16
AVSS_31 1T17
AVSS_32 1U12
AVSS_33 1U13
AVSS_34 1U15
AVSS_35 1U17
AVSS_36 1V7
AVSS_37 1V12
AVSS_38 1V13
AVSS_39 1V18
AVSS1_1 H37
AVSS1_2 L37
AVSS1_3 P37
AVSS1_4 U37
AVSS1_5 Y37
AVSS1_6 1F20
AVSS1_7 1G16
AVSS1_8 1G18
AVSS1_9 1G19
AVSS1_10 1G20
AVSS1_11 1J15
AVSS1_12 1J18
AVSS1_13 1K18
AVSS1_14 1L19

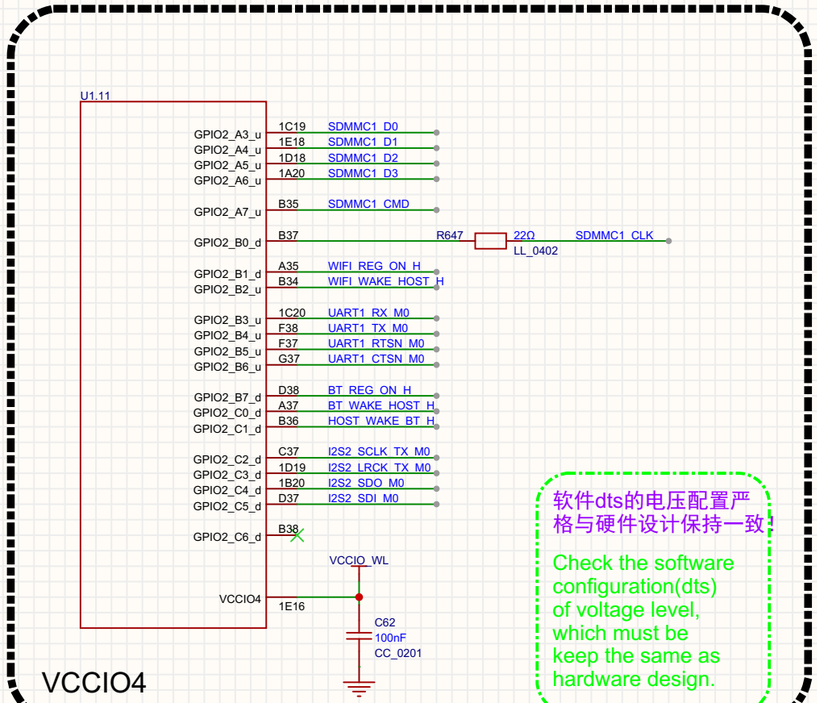

U1.6
LPDDR4_DQ0_A
LPDDR4_DQ1_A
LPDDR4_DQ2_A
LPDDR4_DQ3_A
LPDDR4_DQ4_A
LPDDR4_DQ5_A
LPDDR4_DQ6_A
LPDDR4_DQ7_A
LPDDR4_DMI0_A
LPDDR4_DQS0P_A
LPDDR4_DQS0N_A
LPDDR4_DQ8_A
LPDDR4_DQ9_A
LPDDR4_DQ10_A
LPDDR4_DQ11_A
LPDDR4_DQ12_A
LPDDR4_DQ13_A
LPDDR4_DQ14_A
LPDDR4_DQ15_A
LPDDR4_DMI1_A
LPDDR4_DQS1P_A
LPDDR4_DQS1N_A
LPDDR4_DQ0_B
LPDDR4_DQ1_B
LPDDR4_DQ2_B
LPDDR4_DQ3_B
LPDDR4_DQ4_B
LPDDR4_DQ5_B
LPDDR4_DQ6_B
LPDDR4_DQ7_B
LPDDR4_DMI0_B
LPDDR4_DQS0P_B
LPDDR4_DQS0N_B
LPDDR4_DQ8_B
LPDDR4_DQ9_B
LPDDR4_DQ10_B
LPDDR4_DQ11_B
LPDDR4_DQ12_B
LPDDR4_DQ13_B
LPDDR4_DQ14_B
LPDDR4_DQ15_B
LPDDR4_DMI1_B
LPDDR4_DQS1P_B
LPDDR4_DQS1N_B
G2 F2 F1 E1 M2 K1 K2 1E1 1E2 H2 H1 P2 R1 R2 T2 M1 1F2 1F1 1G2 1G1 N1 N2 B12 B10 1A8 1A7 B13 1A10 1B9 1A9 1B8 A13 A12 B19 A19 A20 B20 A15 B15 1C10 B16 1A11 A17 B17
LPDDR4_CLKP_B
1D2
LPDDR4_A1_A
LPDDR4_CKE1_A
LPDDR4_A3_B
LPDDR4_A5_B
LPDDR4_A1_B
LPDDR4_ODT0_CA_B
LPDDR4_ODT0_CA_A
LPDDR4_CLKN_B
LPDDR4_CKE0_B
LPDDR4_A0_A
LPDDR4_A3_A
LPDDR4_A0_B
LPDDR4_A4_A
LPDDR4_A2_A
LPDDR4_A5_A
LPDDR4_CKE1_B
LPDDR4_A2_B
LPDDR4_A4_B
LPDDR4_ODT1_CA_B
LPDDR4_ODT1_CA_A
LPDDR4_CKE0_A
LPDDR4_CLKP_A
LPDDR4_CLKN_A
LPDDR4_CS0n_A
LPDDR4_CS1n_A
LPDDR4_CS1n_B
LPDDR4_CS0n_B
LPDDR4_RESETn
A7 1D2 C1 1C3 1B6 B8 B9 A10 E2 B7 1A3 C2 1B3 A9 1B1 1C1 1A1 1A2 1B7 1A6 1A4 1C2 B4 A5 B5 B3 A3 1A5 1B5 1C7
LPDDR4_CLKP_B
LPDDR4_A1_A
LPDDR4_CKE1_A
LPDDR4_A3_B
LPDDR4_A5_B
LPDDR4_A1_B
LPDDR4_ODT0_CA_B
LPDDR4_ODT0_CA_A
LPDDR4_CLKN_B
LPDDR4_CKE0_B
LPDDR4_A0_A
LPDDR4_A3_A
LPDDR4_A0_B
LPDDR4_A4_A
LPDDR4_A2_A
LPDDR4_A5_A
LPDDR4_CKE1_B
LPDDR4_A2_B
LPDDR4_A4_B
LPDDR4_ODT1_CA_B
LPDDR4_ODT1_CA_A
LPDDR4_CKE0_A
LPDDR4_CLKP_A
LPDDR4_CLKN_A
LPDDR4_CS0n_A
LPDDR4_CS1n_A
LPDDR4_CS1n_B
LPDDR4_CS0n_B
LPDDR4_RESETn
LPDDR4_CLKP_B
LPDDR4_A1_A
LPDDR4_CKE1_A
LPDDR4_A3_B
LPDDR4_A5_B
LPDDR4_A1_B
LPDDR4_ODT0_CA_B
LPDDR4_ODT0_CA_A
LPDDR4_CLKN_B
LPDDR4_CKE0_B
LPDDR4_A0_A
LPDDR4_A3_A
LPDDR4_A0_B
LPDDR4_A4_A
LPDDR4_A2_A
LPDDR4_A5_A
LPDDR4_CKE1_B
LPDDR4_A2_B
LPDDR4_A4_B
LPDDR4_ODT1_CA_B
LPDDR4_ODT1_CA_A
LPDDR4_CKE0_A
LPDDR4_CLKP_A
LPDDR4_CLKN_A
LPDDR4_CS0n_A
LPDDR4_CS1n_A
LPDDR4_CS1n_B
LPDDR4_CS0n_B
LPDDR4_RESETn
LPDDR4_A0_A
LPDDR4_A1_A
LPDDR4_A2_A
LPDDR4_A3_A
LPDDR4_A4_A
LPDDR4_A5_A
LPDDR4_CLKP_A
LPDDR4_CLKN_A
LPDDR4_CKE0_A
LPDDR4_CKE1_A
LPDDR4_CSN0_A
LPDDR4_CSN1_A
LPDDR4_RESET
LPDDR4_ODT0_CA_A
LPDDR4_ODT0_CA_B
LPDDR4_A0_B
LPDDR4_A1_B
LPDDR4_A2_B
LPDDR4_A3_B
LPDDR4_A4_B
LPDDR4_A5_B
LPDDR4_CLKP_B
LPDDR4_CLKN_B
LPDDR4_CKE0_B
LPDDR4_CKE1_B
LPDDR4_CSN0_B
LPDDR4_CSN1_B
LPDDR4_DQ0_B
LPDDR4_DQ1_B
LPDDR4_DQ2_B
LPDDR4_DQ3_B
LPDDR4_DQ4_B
LPDDR4_DQ5_B
LPDDR4_DQ6_B
LPDDR4_DQ7_B
LPDDR4_DQS0P_B
LPDDR4_DQS0N_B
LPDDR4_DQS1P_B
LPDDR4_DQS1N_B
LPDDR4_DQ8_B
LPDDR4_DQ9_B
LPDDR4_DQ10_B
LPDDR4_DQ11_B
LPDDR4_DQ12_B
LPDDR4_DQ13_B
LPDDR4_DQ14_B
LPDDR4_DQ15_B
LPDDR4_DQS0P_A
LPDDR4_DQS0N_A
LPDDR4_DQS1P_A
LPDDR4_DQS1N_A
LPDDR4_DQ8_A
LPDDR4_DQ9_A
LPDDR4_DQ10_A
LPDDR4_DQ11_A
LPDDR4_DQ12_A
LPDDR4_DQ13_A
LPDDR4_DQ14_A
LPDDR4_DQ15_A
LPDDR4_DQ0_A
LPDDR4_DQ1_A
LPDDR4_DQ2_A
LPDDR4_DQ3_A
LPDDR4_DQ4_A
LPDDR4_DQ5_A
LPDDR4_DQ6_A
LPDDR4_DQ7_A
LPDDR4_DMI0_A
LPDDR4_DMI1_A
LPDDR4_DMI0_B
LPDDR4_DMI1_B
DDR_RZQ
ODR_VREFOUT
DDR_VDDQ_1
DDR_VDDQ_2
DDR_VDDQ_3
DDR_VDDQ_4
DDR_VDDQ_5
DDR_VDDQ_6
DDR_VDDQ_7
DDR_VDDQ_8
DDR_VDDQL_1
DDR_VDDQL_2
DDR_VDDQL_3
DDR_VDDQL_4
DDR_VDDQL_5
DDR_AVSS
1F3 1H3 1D5 1D6 1D7 1D8 1D9 1E4 1F4 1G4 1E6 1E7 1E8 1F8 1G8 1D4
R662
120Ω
VDD2_DDR
C905 100nF 0201
C906 100nF 0201
C907 100nF 0201
C899 4.7uF 0402
C908 10uF 0603
VDDQ_DRAM
C902 100nF 0201
C903 100nF 0201
C904 100nF 0201
C900 4.7uF 0402
C901 4.7uF 0402
VDDQ VDDQL
LPDDR4 1.10V 1.10V
LPDDR4X 1.10V 0.60V


U1.7
R2
22Ω
LL_0402
XOUT24M
AD38
XOUT24M
R3
1MΩ
LL_0402
X6
24MHz
XIN24M
AD37
XIN24M
C909
12pF
0402
C910
12pF
0402
R636
0Ω
RR_0201
1M18
TVSS
1N19
GPIO0_D3_d
1M19
GPIO0_D4_d
1N20
GPIO0_D5_d
1P20
GPIO0_D6_d
VDDA0V9_PMU
C911
1uF
0402
C914
100nF
0201
1P17
PMUPLL_AVDD_0V9
VCCA1V8_PMU
C912
1uF
0402
C41
100nF
0201
1N18
PMUPLL_AVDD_1V9
1N17
PMUPLL_AVSS
VDDA_0V9
C42
1uF
0201
C915
100nF
0201
1H5
SYSPLL_AVDD_0V9
VCCA_1V8
C913
1uF
0402
C916
100nF
0201
1H4
SYSPLL_AVDD_1V8
1J5
SYSPLL_AVSS
nPOR_u
AG38
RESETN
C917
100nF
CC_0201
R4
0Ω
GPIO0_A0_d
1P18
GPIO0_A1_z
1R20
TSADC_SEUT_M0
GPIO0_A2_d
1P19
PMIC_SLEEP_H
GPIO0_A3_u
1R19
PMIC_INT_L
GPIO0_A4_u
AF37
SDMMC0_DET_L
GPIO0_A5_d
AG37
GPIO0_A5
PWR_EN
GPIO0_A6_d
AJ38
GPIO0_A7_u
AF38
FLASH_VOL_SEL
VCC3V3_PMU
PMUIO1
1P16
C918
100nF
0201
GPIO0_B0_u
1T20
DVP_PWREN_H
GPIO0_B1_u
AJ37
I2C0_SCL_PMIC
GPIO0_B2_u
AK38
I2C0_SDA_PMIC
GPIO0_B3_u
AK37
I2C1_SCL_TP
I2C_SDA
GPIO0_B4_u
AM38
I2C1_SDA_TP
I2C_SCL
GPIO0_B5_u
1U20
GPIO0_B6_u
1R16
GPIO0_B7_d
1R17
GPIO0_C0_d
1T19
GPIO0_C1_d
1R18
GPIO0_C2_d
AM37
GPIO0_C3_d
AN38
GPIO0_C4_d
AN37
GPIO0_C5_d
1U19
GPIO0_C5
PWM6_LCD_BL
GPIO0_C6_d
1T18
GPIO0_C6
PWM_FAN
GPIO0_C7_d
1V20
GPIO0_C7
KEY_VOLUME-
GPIO0_D0_u
1V19
UART2_RX_M0_DEBUG
GPIO0_D1_u
1U18
UART2_TX_M0_DEBUG
VCC3V3_PMU
PMUIO2
1N15
C919
100nF
0201
VDDA0V9_PMU
PMU_VDD_LOGIC_0V9
1N16
C920
100nF
0201
C48
1uF
0402
VCC3V3_PMU
R6
I2C0_SCL_PMIC
LL_0402 2.2kΩ ±1% 62.5mW
R637
I2C0_SDA_PMIC
LL_0402 2.2kΩ ±1% 62.5mW
VCC3V3_PMU
R638
I2C1_SCL_TP
LL_0402
2.2kΩ ±1% 62.5mW
R639
I2C1_SDA_TP
LL_0402
2.2kΩ ±1% 62.5mW


U1.9
GPIO1_B4_u
GPIO1_B5_u
GPIO1_B6_u
GPIO1_B7_u
GPIO1_C0_u
GPIO1_C1_u
GPIO1_C2_u
GPIO1_C3_u
GPIO1_C4_u
GPIO1_C5_d
GPIO1_C6_d
GPIO1_C7_d
GPIO1_D0_d
GPIO1_D1_u
GPIO1_D2_u
GPIO1_D3_u
GPIO1_D4_u
A32 EMMC_D0
B27 EMMC_D1
B32 EMMC_D2
B29 EMMC_D3
B33 EMMC_D4
A30 EMMC_D5
B30 EMMC_D6
A33 EMMC_D7
A27 EMMC_CMD
A29 R641 22Ω EMMC_CLKOUT
1A16 EMMC_DATA_STROBE/FLASH_CLE
1B16 EMMC_RSTN
1A15
1A17
1A18
1B17
1C15
VCCIO2
1C13
VCCIO_FLASH
C49
CC_0201
100nF
EMMC_DATA_STROBE/FLASH_CLE
R640
0Ω
EMMC_DATA_STROBE
eMMC Short 'eMMC_D0' and
GND to enter Maskrom Mode.
Be sure to reserve the testpoints for
firmware burning or update.
U1.10
GPIO1_D5_u
GPIO1_D6_u
GPIO1_D7_u
GPIO2_A0_u
GPIO2_A1_u
GPIO2_A2_d
VCCIO3
1E20 SDMMC0_D0
1F19 SDMMC0_D1
1D20 SDMMC0_D2
1F18 SDMMC0_D3
1E19 SDMMC0_CMD
G38 R14 22Ω SDMMC0_CLK
1F17
VCCIO_SD
C942
CC_0201
100nF
软件dts的电压配置严格与硬件设计保持一致！
Check the software configuration(dts)
of voltage level, which must be
keep the same as hardware design.
VCCIO2/3

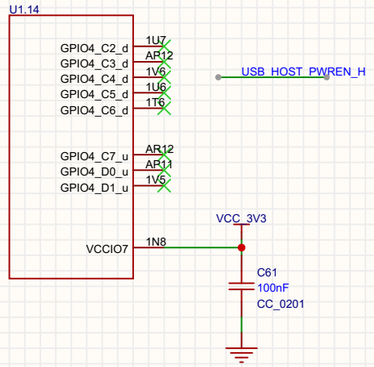

U1.21
USB_OTG0_DP
USB_OTG0_DM
USB_OTG0_VBUSDET
USB_OTG0_ID
R37 USB_OTG0_DP
R38 USB_OTG0_DM
T38 USB_OTG0_VBUSDET
T37
C928
100nF
0402
USB_HOST1_DP
USB_HOST1_DM
1J19 USB3_HOST1_DP
1J20 USB3_HOST1_DM
DPU
DMU
VDDA_0V9
USB_AVDD1_0V9
1J17
VCCA_1V8
USB_AVDD1_1V9
1H16
VCC_3V3
USB_AVDD1_3V3
1H18
C921
100nF
CC_0201
C922
100nF
CC_0201
C923
100nF
CC_0201
USB3_HOST1_SSTXp/SATA1_TXp
USB3_HOST1_SSTXn/SATA1_TXn
W37
W38
USB3_HOST1_SSRXp/SATA1_RXp
USB3_HOST1_SSRXn/SATA1_RXx
V38
V37
PCIE20_TXp/SATA2_TXp
PCIE20_TXn/SATA2_TXn
AA37
AA38
PCIE20_RXp/SATA2_RXp
PCIE20_RXn/SATA2_RXn
AB37
AB38
PCIE20_REFCLKp
PCIE20_REFCLKn
1K19
1K20
VDDA_0V9
MULTI_PHY_AVDD_0V9
1K16
VCCA_1V8
MULTI_PHY_AVDD_1V8
1K17
C950
100nF
CC_0201
C59
100nF
0402
C927
100nF
CC_0201
C60
4.7uF
0402
USB3.0/PCIe2.0x1/SATA


U1.22
USB_HOST2_DP
USB_HOST2_DM
V2 USB2_HOST2_DP
V1 USB2_HOST2_DM
USB_HOST3_DP
USB_HOST3_DM
Y2
Y1
USB_AVDD2_0V9
1J4
USB_AVDD2_1V8
1J3
USB_AVDD2_3V3
1J2
Caps should be placed
under the U20 package
USB2.0 HOST



软件dts的电压配置严格与硬件设计保持一致！

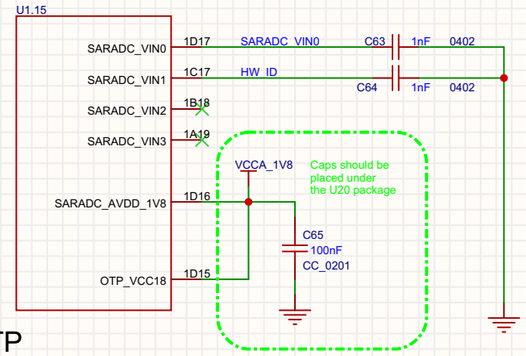
Check the software configuration(dts) of voltage level, which must be keep the same as hardware design.

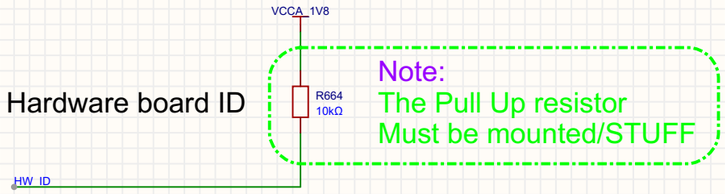


VCCIO7

软件dts的电压配置严格与硬件设计保持一致！

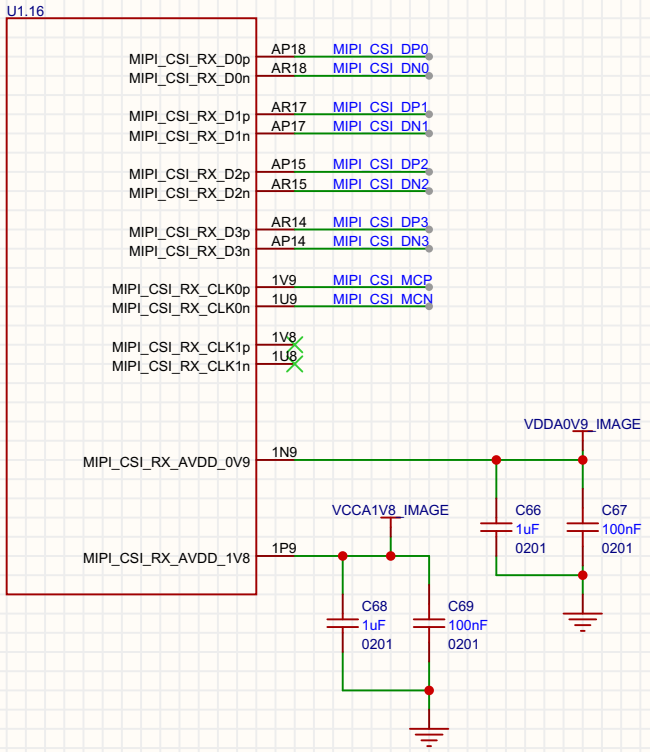
Check the software configuration(dts) of voltage level, which must be keep the same as hardware design.



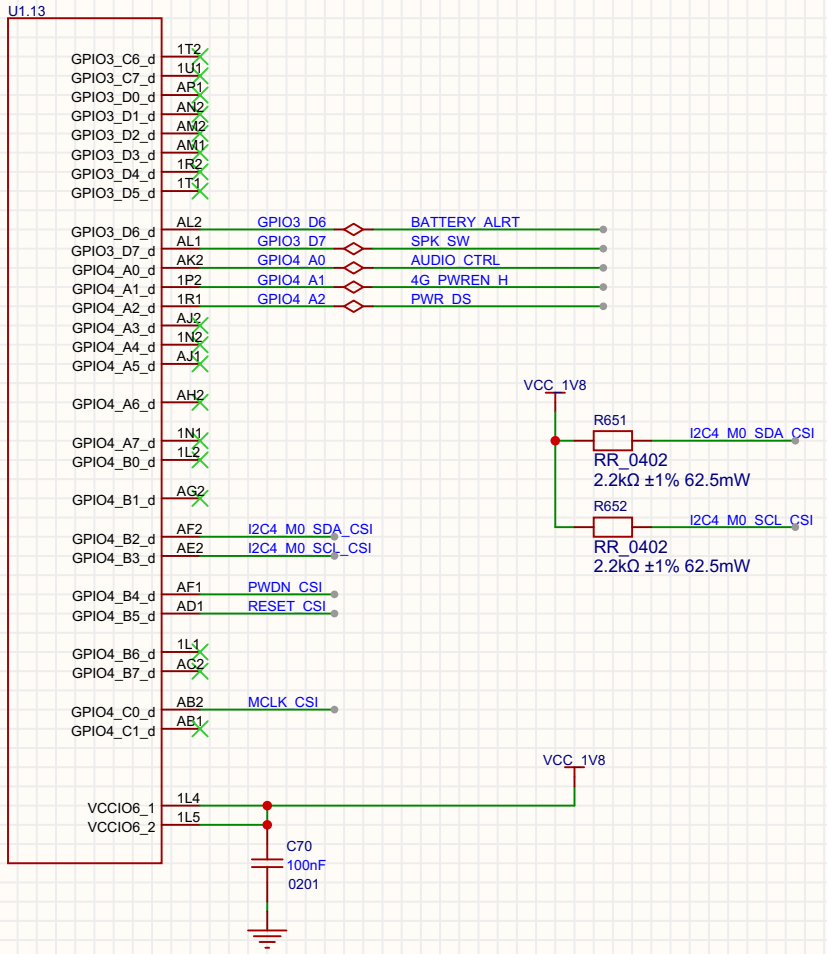


SARADC/OTP

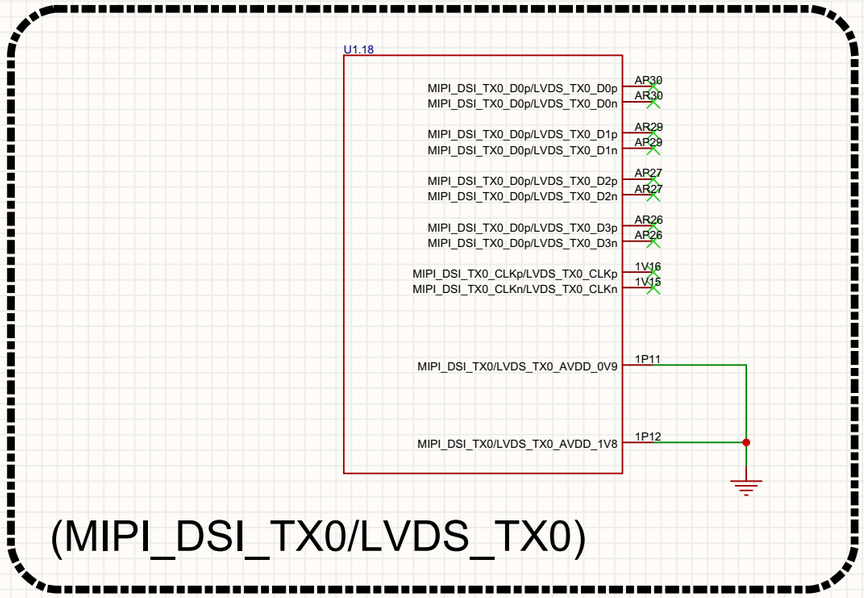
软件dts的电压配置严格与硬件设计保持一致！
Check the software configuration(dts)
of voltage level, which must be
keep the same as hardware design.

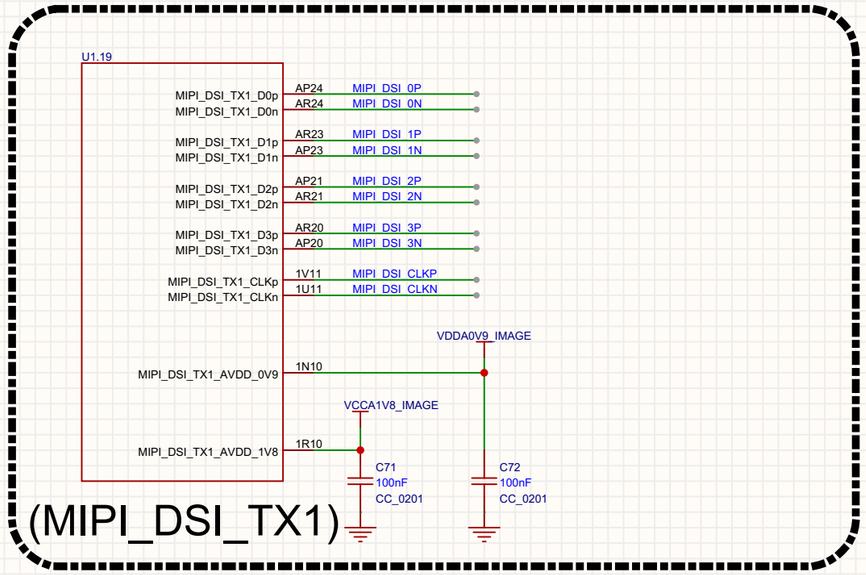


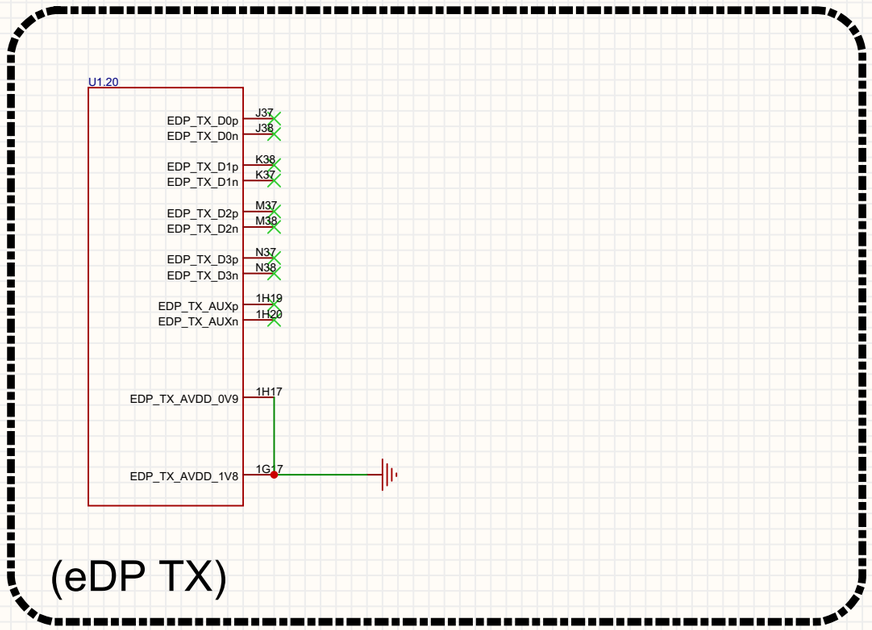
MIPI_CSI_RX

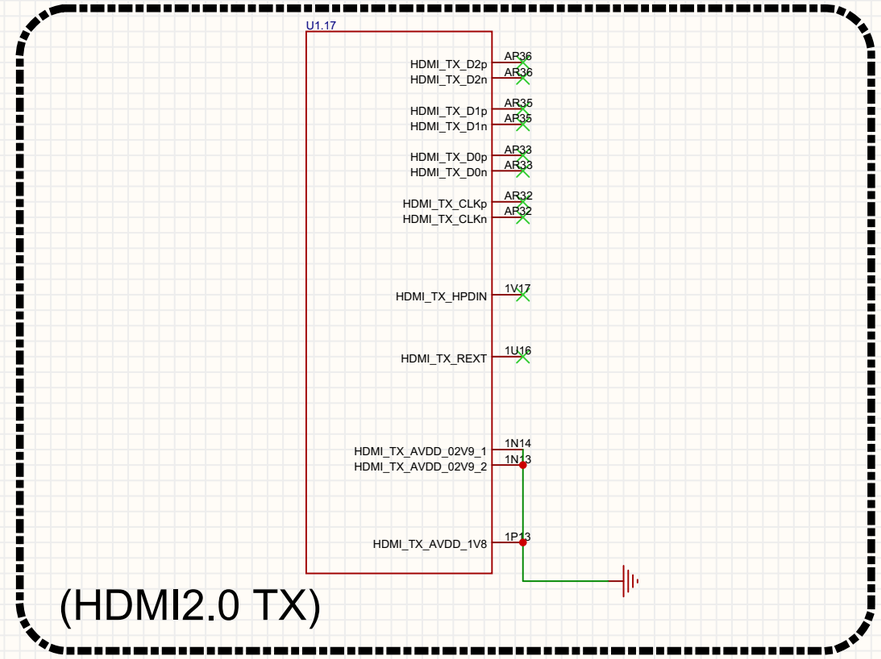


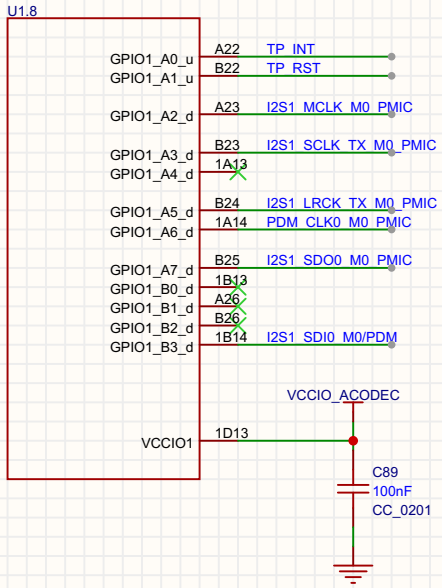
VCCIO6


U1.18
MIPI_DSI_TX0_D0p/LVDS_TX0_D0p
MIPI_DSI_TX0_D0p/LVDS_TX0_D0n
MIPI_DSI_TX0_D0p/LVDS_TX0_D1p
MIPI_DSI_TX0_D0p/LVDS_TX0_D1n
MIPI_DSI_TX0_D0p/LVDS_TX0_D2p
MIPI_DSI_TX0_D0p/LVDS_TX0_D2n
MIPI_DSI_TX0_D0p/LVDS_TX0_D3p
MIPI_DSI_TX0_D0p/LVDS_TX0_D3n
MIPI_DSI_TX0_CLKp/LVDS_TX0_CLKp
MIPI_DSI_TX0_CLKn/LVDS_TX0_CLKn
MIPI_DSI_TX0/LVDS_TX0_AVDD_0V9
MIPI_DSI_TX0/LVDS_TX0_AVDD_1V8
AR30
AR30
AR29
AR29
AR27
AR27
AR26
AR26
1V16
1V15
1P11
1P12
(MIPI_DSI_TX0/LVDS_TX0)


U1.19
MIPI_DSI_TX1_D0p
MIPI_DSI_TX1_D0n
MIPI_DSI_TX1_D1p
MIPI_DSI_TX1_D1n
MIPI_DSI_TX1_D2p
MIPI_DSI_TX1_D2n
MIPI_DSI_TX1_D3p
MIPI_DSI_TX1_D3n
MIPI_DSI_TX1_CLKp
MIPI_DSI_TX1_CLKn
MIPI_DSI_TX1_AVDD_0V9
MIPI_DSI_TX1_AVDD_1V8
AP24
AR24
AR23
AP23
AP21
AR21
AR20
AP20
1V11
1U11
1N10
1R10
MIPI_DSI_0P
MIPI_DSI_0N
MIPI_DSI_1P
MIPI_DSI_1N
MIPI_DSI_2P
MIPI_DSI_2N
MIPI_DSI_3P
MIPI_DSI_3N
MIPI_DSI_CLKP
MIPI_DSI_CLKN
VDDA0V9_IMAGE
VCCA1V8_IMAGE
C71
100nF
CC_0201
C72
100nF
CC_0201
(MIPI_DSI_TX1)


U1.20
EDP_TX_D0p
EDP_TX_D0n
EDP_TX_D1p
EDP_TX_D1n
EDP_TX_D2p
EDP_TX_D2n
EDP_TX_D3p
EDP_TX_D3n
EDP_TX_AUXp
EDP_TX_AUXn
EDP_TX_AVDD_0V9
EDP_TX_AVDD_1V8
J37
J36
K38
K37
M37
M38
N37
N38
1H19
1H20
1H17
1G17
(eDP TX)

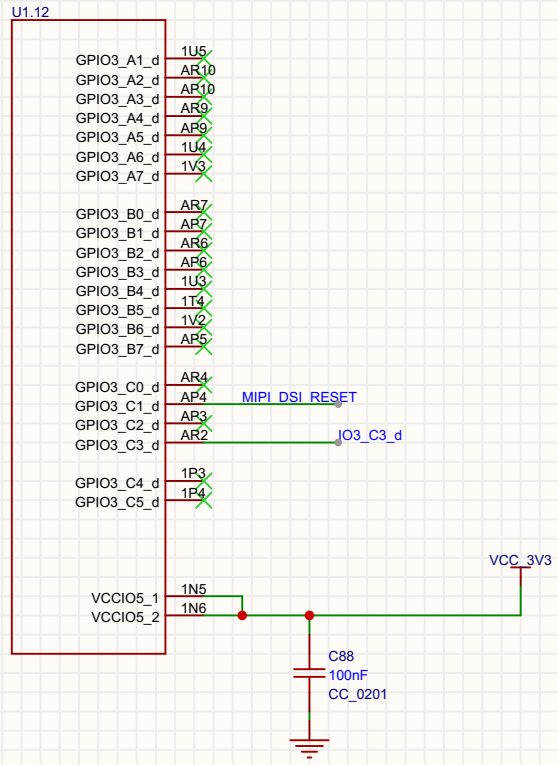

U1.17
HDMI_TX_D2p
HDMI_TX_D2n
HDMI_TX_D1p
HDMI_TX_D1n
HDMI_TX_D0p
HDMI_TX_D0n
HDMI_TX_CLKp
HDMI_TX_CLKn
HDMI_TX_HPDIN
HDMI_TX_REXT
HDMI_TX_AVDD_02V9_1
HDMI_TX_AVDD_02V9_2
HDMI_TX_AVDD_1V8
AP36
AR36
AR35
AR35
AR33
AR33
AR32
AR32
1V17
1U16
1N14
1N13
1P13
(HDMI2.0 TX)



软件dts的电压配置严格与硬件设计保持一致！

Check the software configuration(dts)
of voltage level, which must be
keep the same as hardware design.

VCCIO1

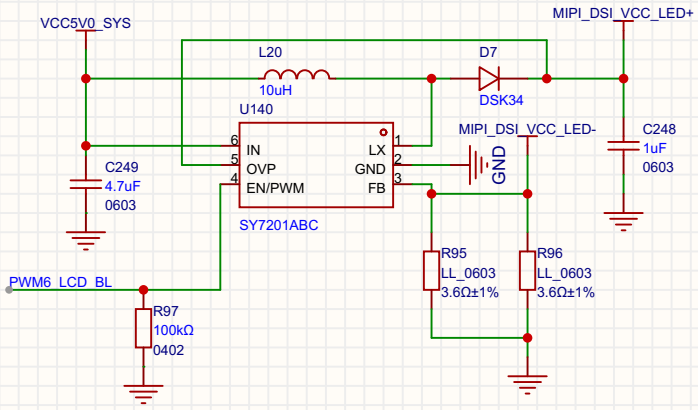


软件dts的电压配置严格与硬件设计保持一致！

Check the software configuration(dts)
of voltage level, which must be
keep the same as hardware design.

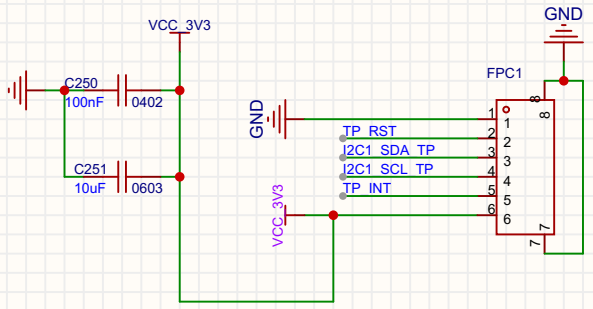
VCCIO5

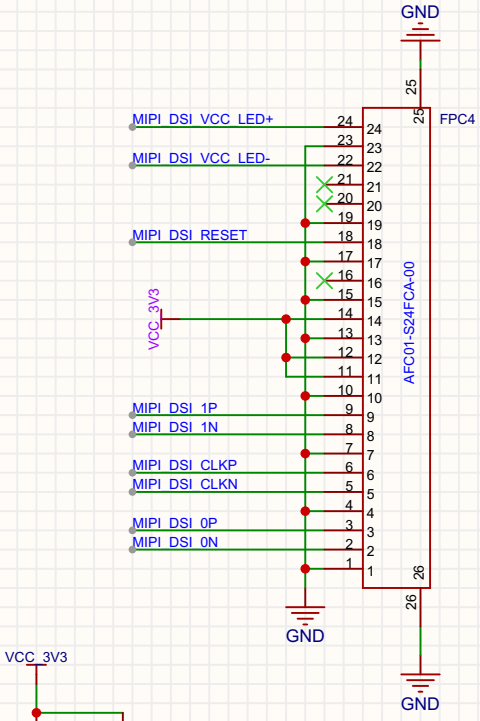
LCM_MIPI_DSI_Panel

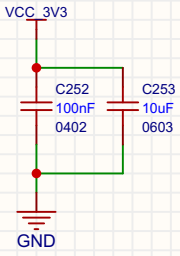


Adjust power supply according to specific panel needs.

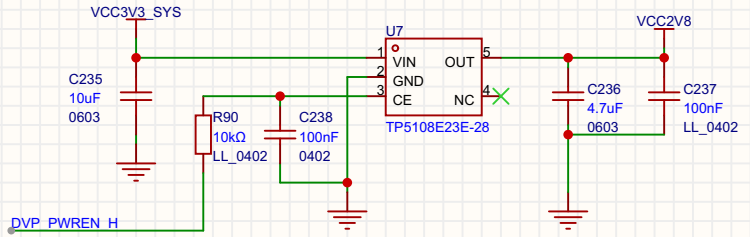
TP



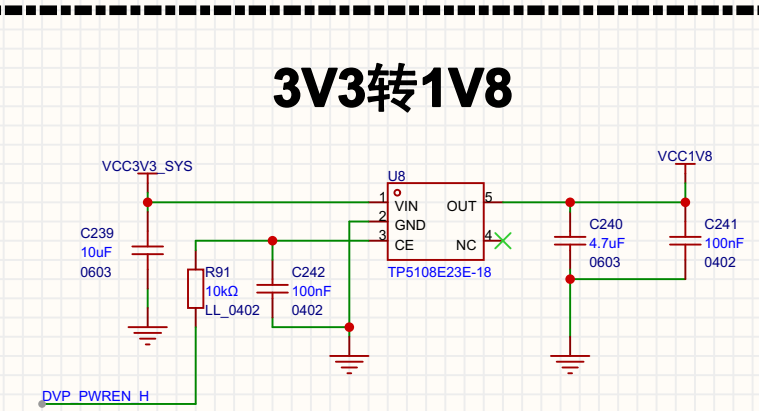




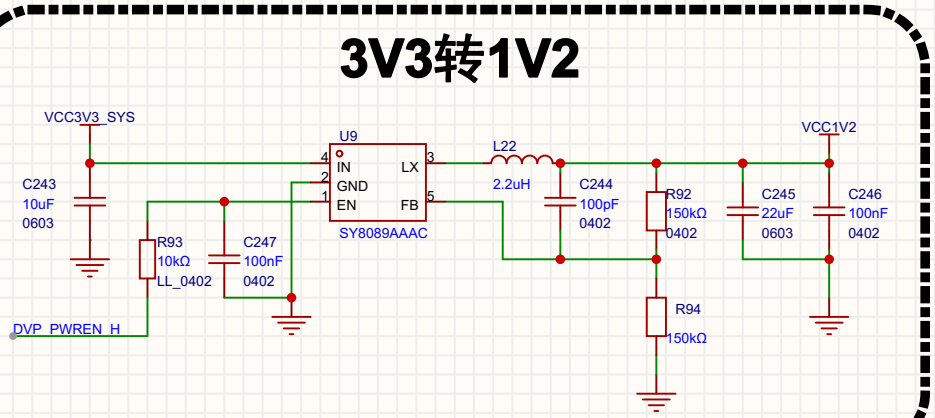
3V3转2V8

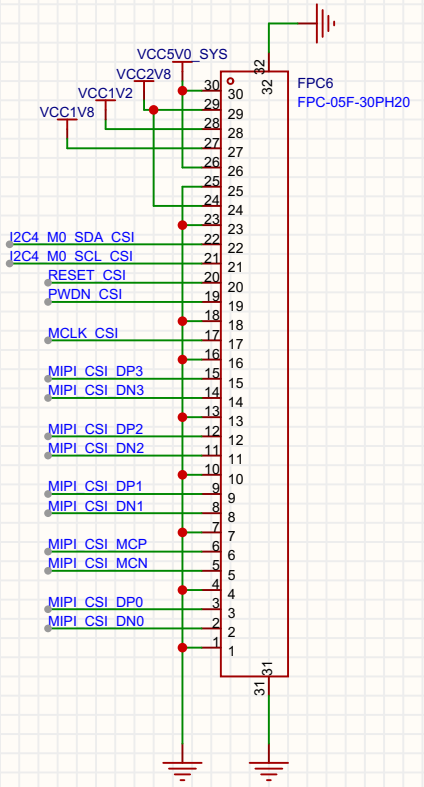


3V3转1V8



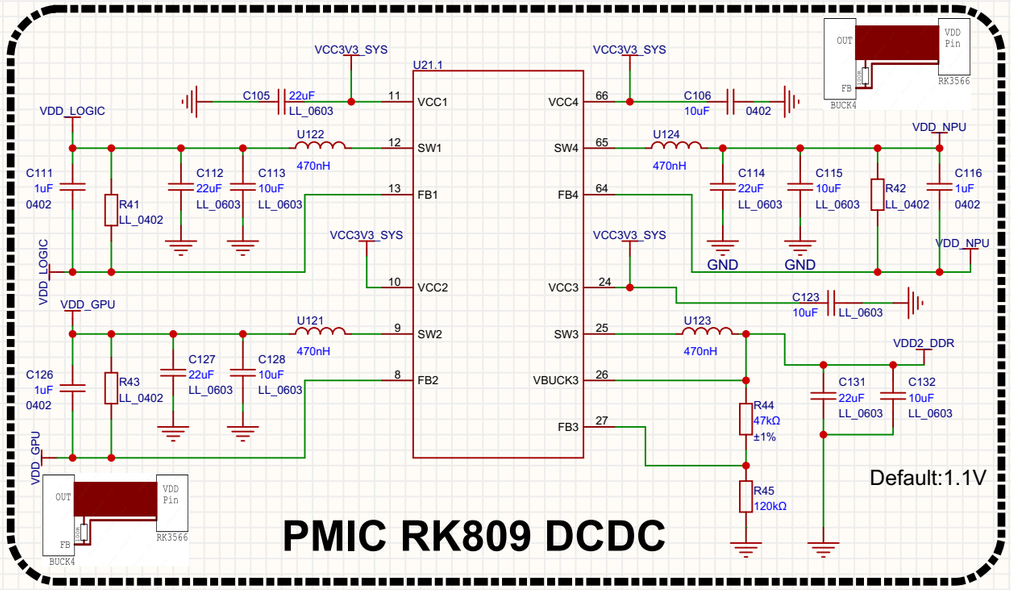
3V3转1V2

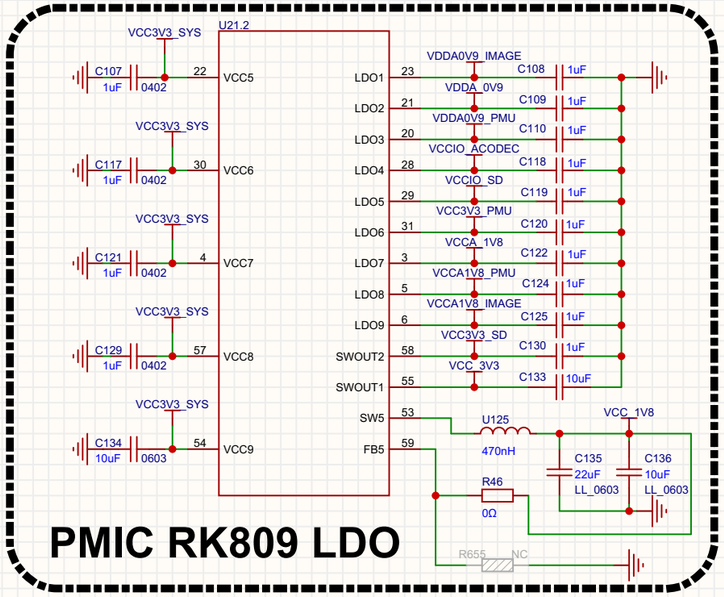


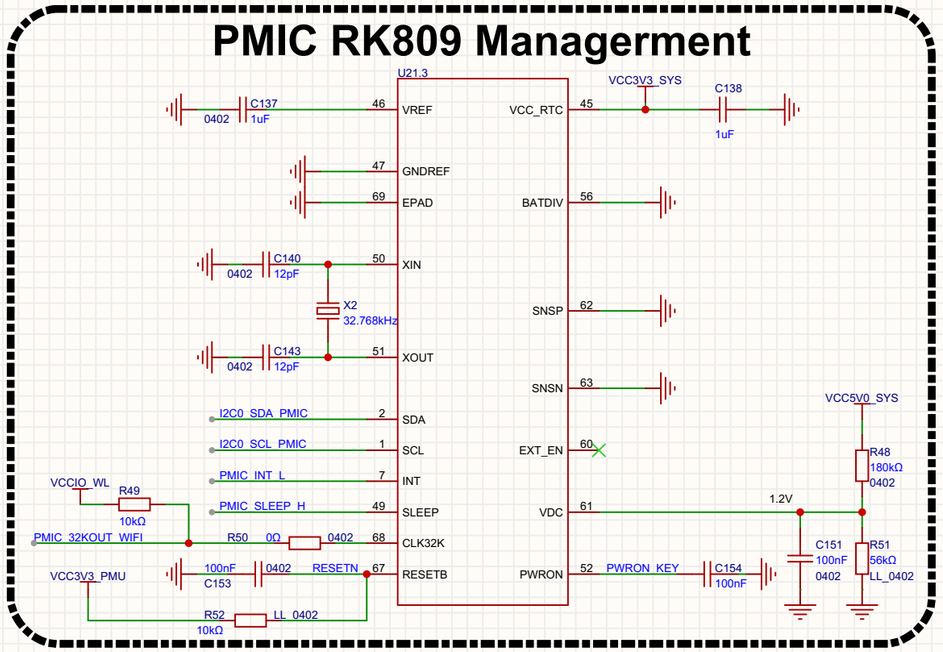


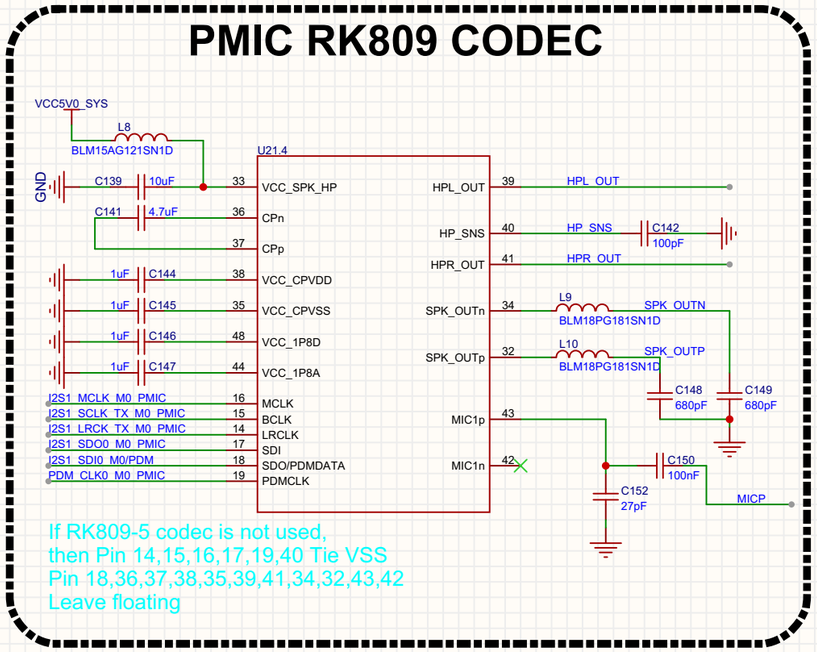
MIPI Camera

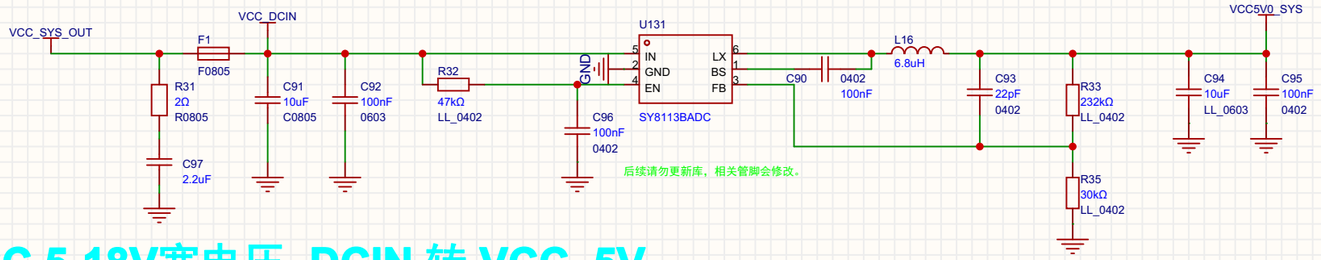
(MIPI_CSI_RX 4lanes)


VCC3V3_SYS
U21.1
VDD_LOGIC
C105 22uF LL_0603
C111 1uF 0402
R41 LL_0402
C112 22uF LL_0603
C113 10uF LL_0603
U122 470nH
VCC1
SW1
FB1
VCC2
VDD_GPU
U121 470nH
C126 1uF 0402
R43 LL_0402
C127 22uF LL_0603
C128 10uF LL_0603
SW2
FB2
VCC4
C106 10uF 0402
SW4
U124 470nH
C114 22uF LL_0603
C115 10uF LL_0603
R42 LL_0402
C116 1uF 0402
VDD_NPU
FB4
GND
VCC3
SW3
U123 470nH
C123 10uF LL_0603
VDD2_DDR
C131 22uF LL_0603
C132 10uF LL_0603
VBUCK3
FB3
R44 47kΩ ±1%
R45 120kΩ
Default:1.1V
BUCK4
RK3566
PMIC RK809 DCDC


U21.2
VCC3V3_SYS
C107 1uF 0402
VCC5
C117 1uF 0402
VCC6
C121 1uF 0402
VCC7
C129 1uF 0402
VCC8
C134 10uF 0603
VCC9
LDO1
LDO2
LDO3
LDO4
LDO5
LDO6
LDO7
LDO8
LDO9
SWOUT2
SWOUT1
SW5
FB5
VDDA0V9_IMAGE
VDDA_0V9
VDDA0V9_PMU
VCCIO_ACODEC
VCCIO_SD
VCC3V3_PMU
VCCA_1V8
VCCA1V8_PMU
VCCA1V8_IMAGE
VCC3V3_SD
VCC_3V3
C108 1uF
C109 1uF
C110 1uF
C118 1uF
C119 1uF
C120 1uF
C122 1uF
C124 1uF
C125 1uF
C130 1uF
C133 10uF
U125 470nH
VCC_1V8
C135 22uF LL_0603
C136 10uF LL_0603
R46 0Ω
R655 NC
PMIC RK809 LDO

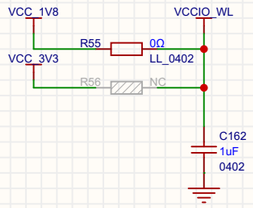

U21.3
C137 1uF 0402
VREF
GNDREF
EPAD
C140 12pF 0402
XIN
X2 32.768kHz
C143 12pF 0402
XOUT
I2C0_SDA_PMIC
SDA
I2C0_SCL_PMIC
SCL
PMIC_INT_L
INT
PMIC_SLEEP_H
SLEEP
PMIC_32KOUT_WIFI
R50 0Ω 0402
CLK32K
VCC3V3_PMU
100nF C153 0402
RESETN
RESETB
VCCIO_WL
R49 10kΩ
R52 10kΩ
LL_0402
VCC_RTC
C138 1uF
BATDIV
SNSP
SNSN
EXT_EN
VDC
PWRON
PWRON_KEY
C154 100nF
VCC5V0_SYS
R48 180kΩ 0402
1.2V
C151 100nF 0402
R51 56kΩ LL_0402
PMIC RK809 Managerment

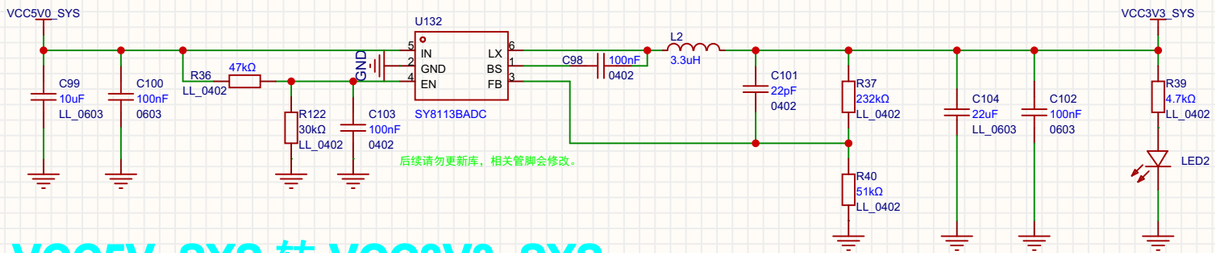

VCC5V0_SYS
L8
BLM15AG121SN1D
U21.4
GND
C139 10uF
C141 4.7uF
VCC_SPK_HP
CPn
CPp
VCC_CPVDD
VCC_CPVSS
VCC_1P8D
VCC_1P8A
I2S1_MCLK_M0_PMIC
I2S1_SCLK_TX_M0_PMIC
I2S1_LRCK_TX_M0_PMIC
I2S1_SDO0_M0_PMIC
I2S1_SDI0_M0/PDM
PDM_CLK0_M0_PMIC
MCLK
BCLK
LRCLK
SDI
SDO/PDMDATA
PDMCLK
HPL_OUT
HP_SNS
HPR_OUT
SPK_OUTn
SPK_OUTp
MIC1p
MIC1n
C142 100pF
L9
BLM18PG181SN1D
L10
C148 680pF
C149 680pF
C150 100nF
C152 27pF
MICP
If RK809-5 codec is not used,
then Pin 14,15,16,17,19,40 Tie VSS
Pin 18,36,37,38,35,39,41,34,32,43,42
Leave floating
PMIC RK809 CODEC



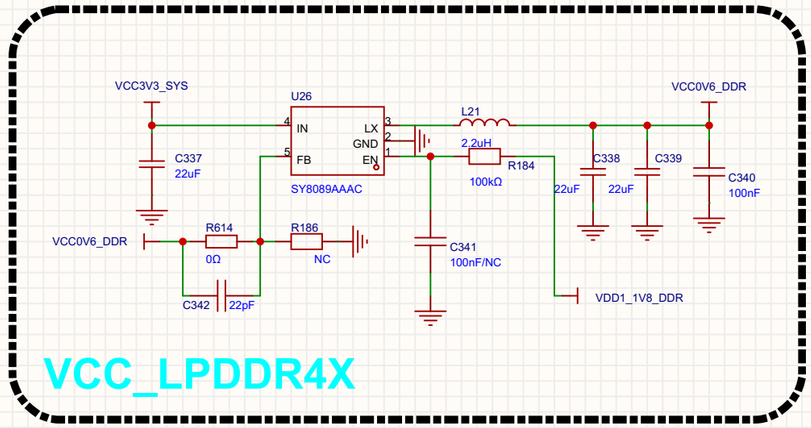
VCC 5-18V宽电压_DCIN 转 VCC_5V

VCCIO_WL

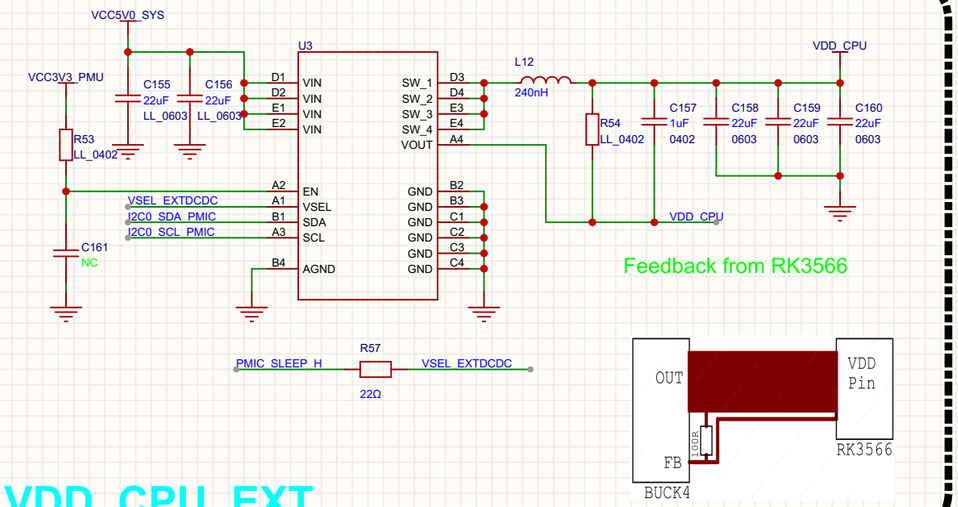




VCC5V_SYS 转 VCC3V3_SYS

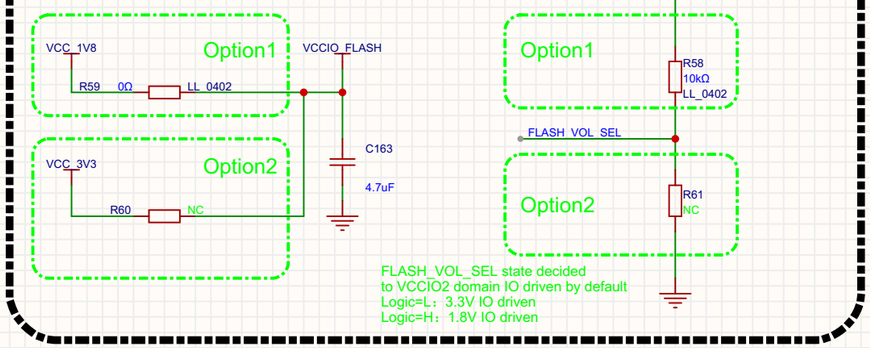


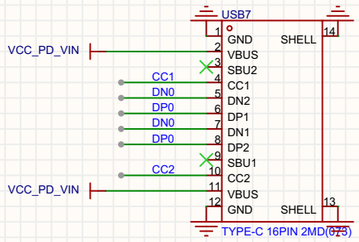
VCC_LPDDR4X



VDD_CPU_EXT

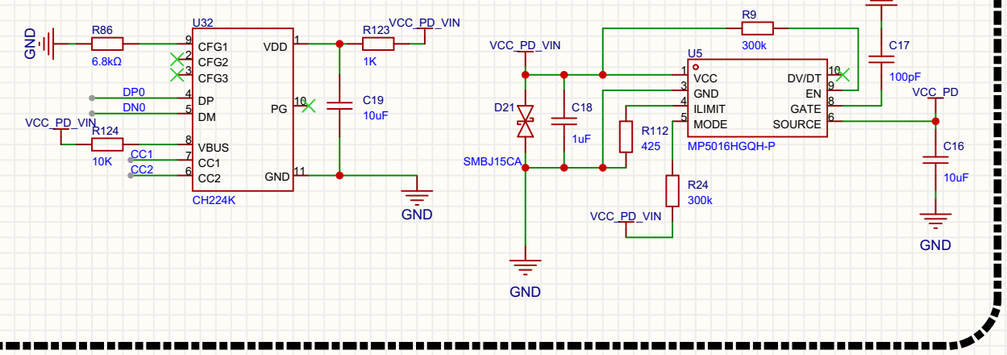
Flash Power Manage

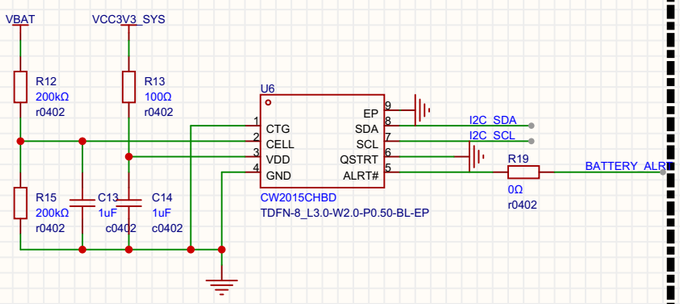




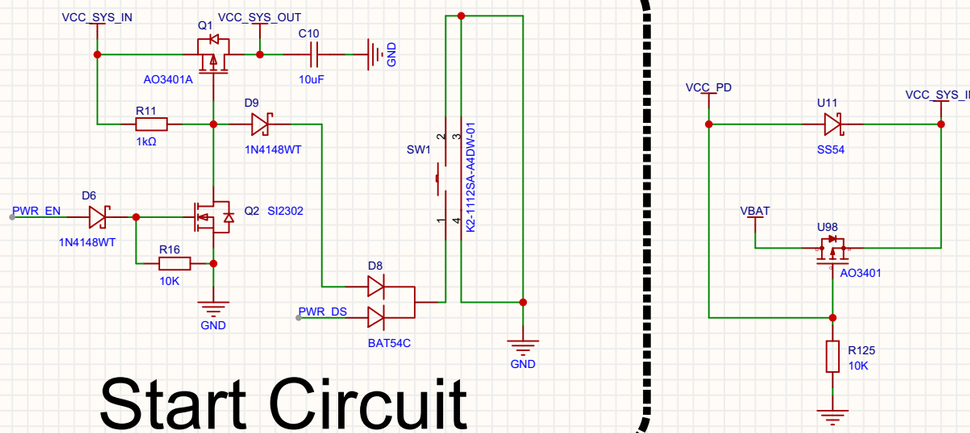
$$I_{LIMIT}(A) = \frac{0.55(V)}{R_{LIMIT}(\Omega)} \times 3870$$

POWER PD 9V MAX 5A

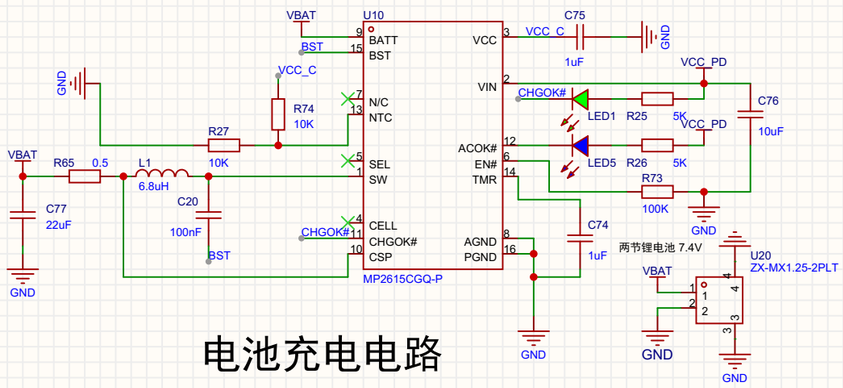




BATTERY MONITOR



Start Circuit

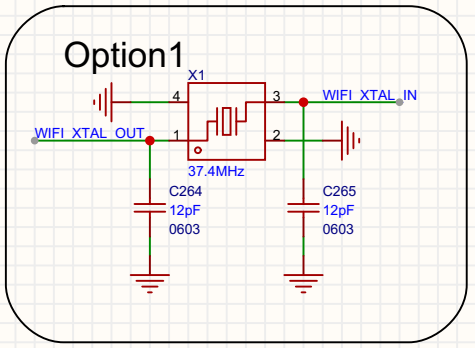


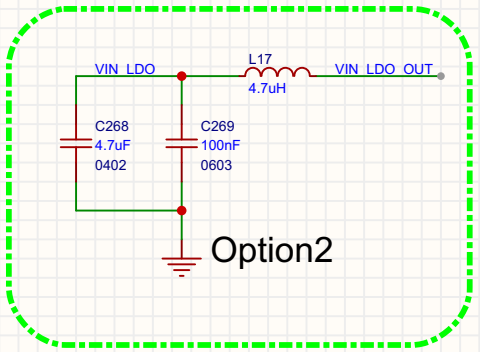
电池充电电路

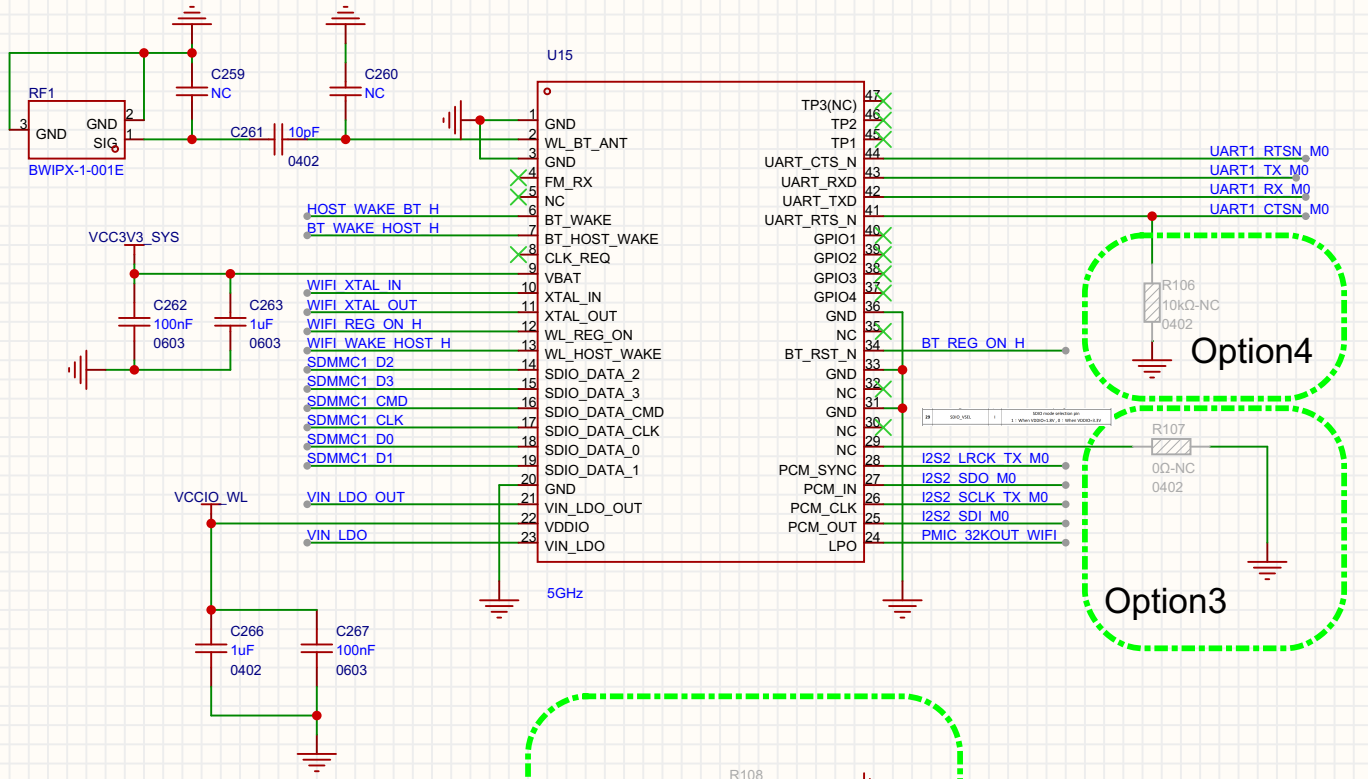
电源路径选择

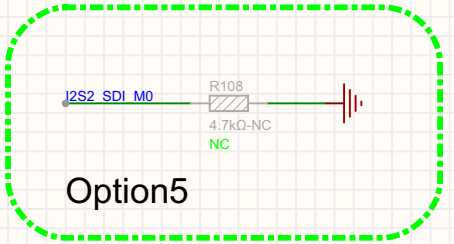
原理图	RK3566_2	创建日期	2026-09-20
		更新日期	2026-09-20
板子	RK3566手机V1.0	图页	Power_Charge
绘制		RK3566手机	
审阅			
		版本	尺寸
嘉立创EDA		V1.0	A4
		页 15 共 25	
		嘉立创EDA	

SDIO WIFI/BT

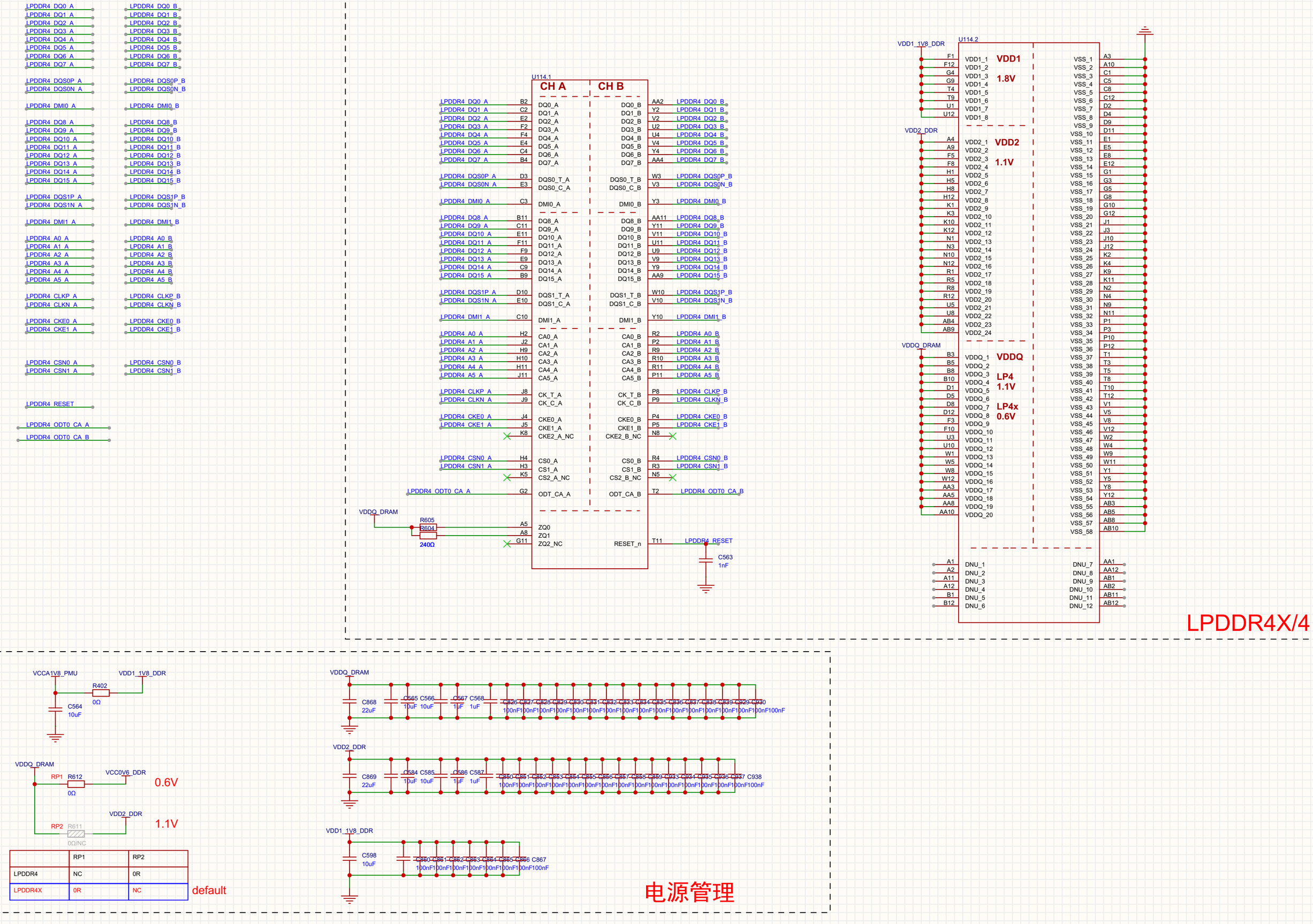


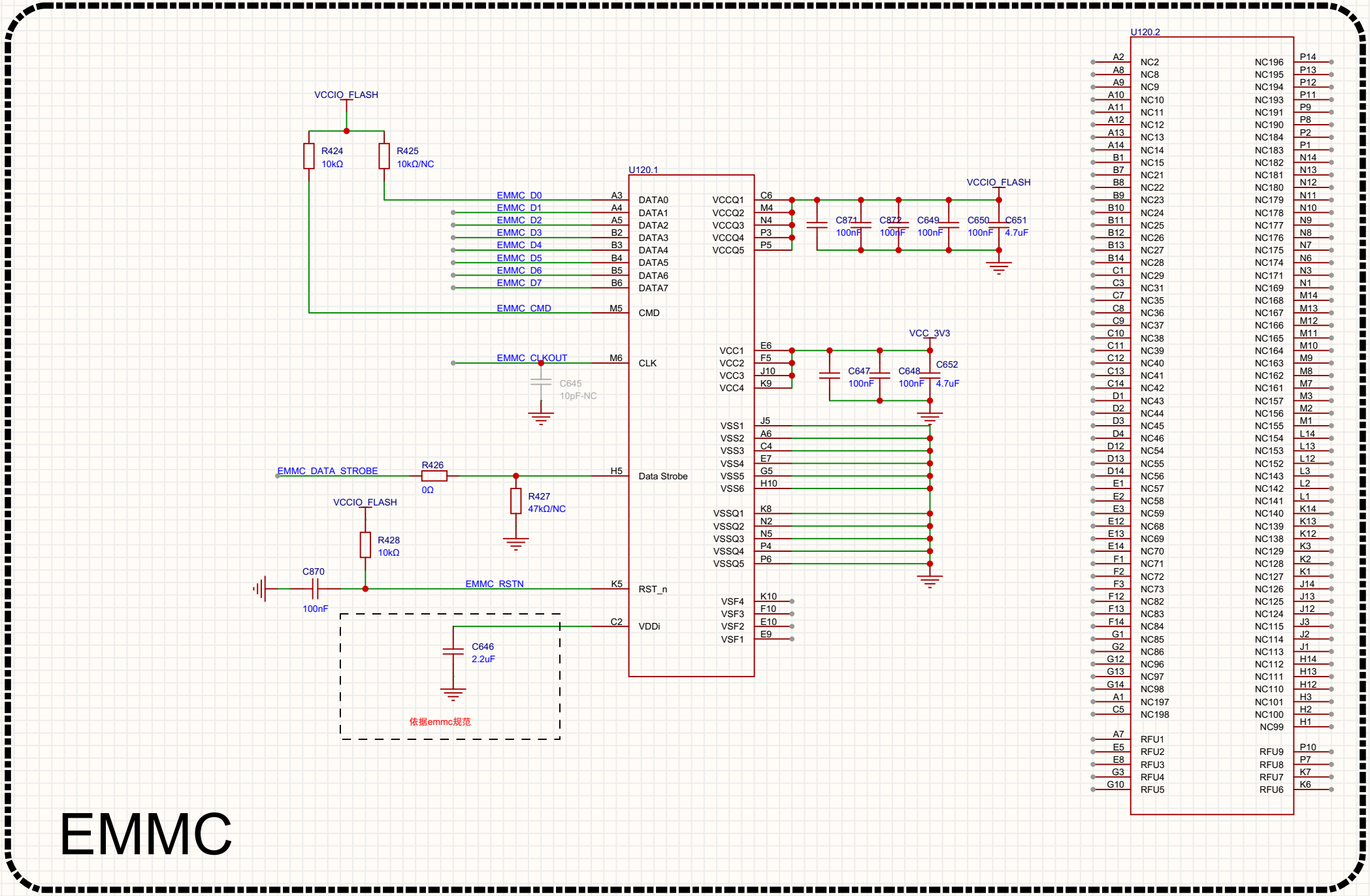


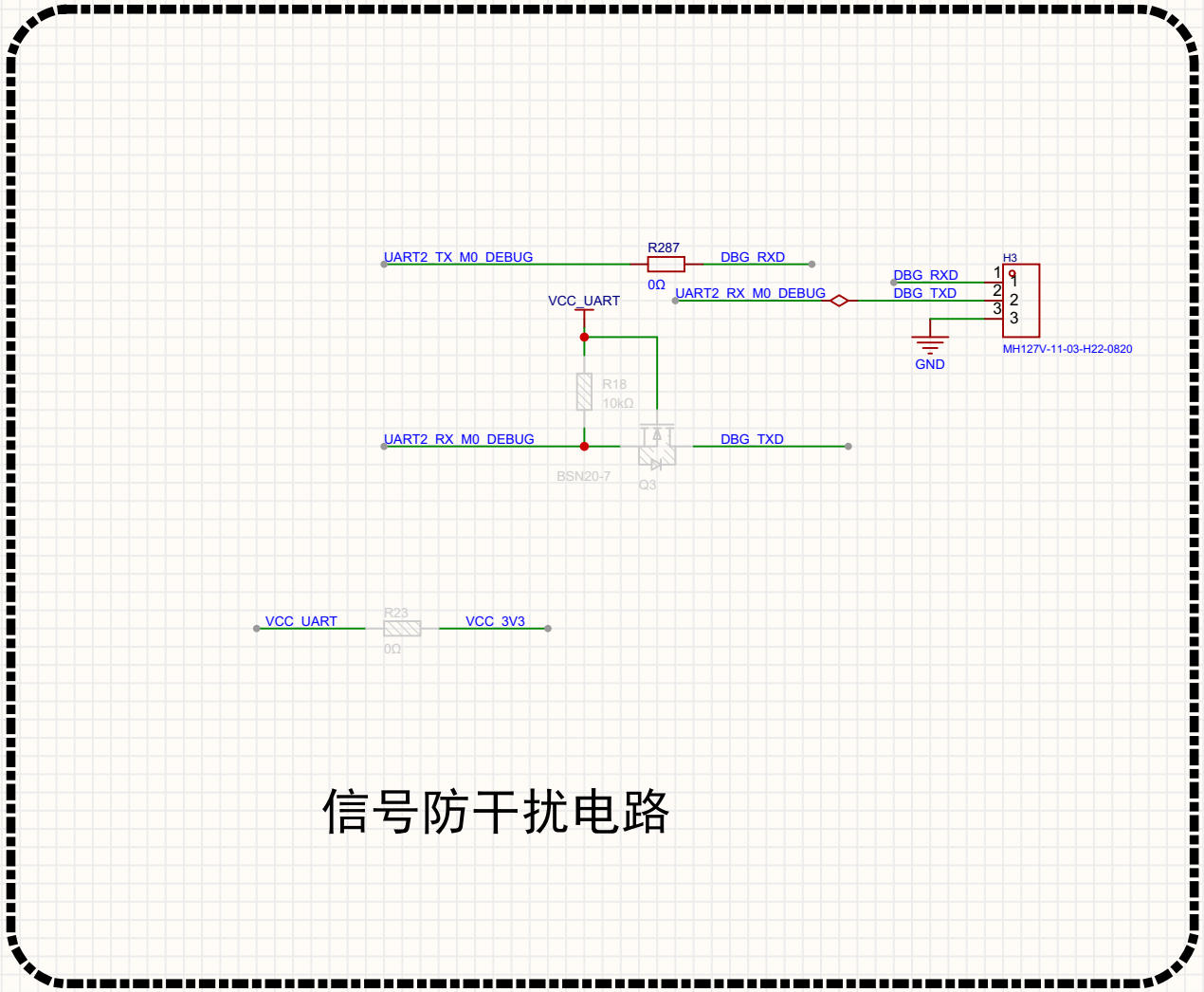


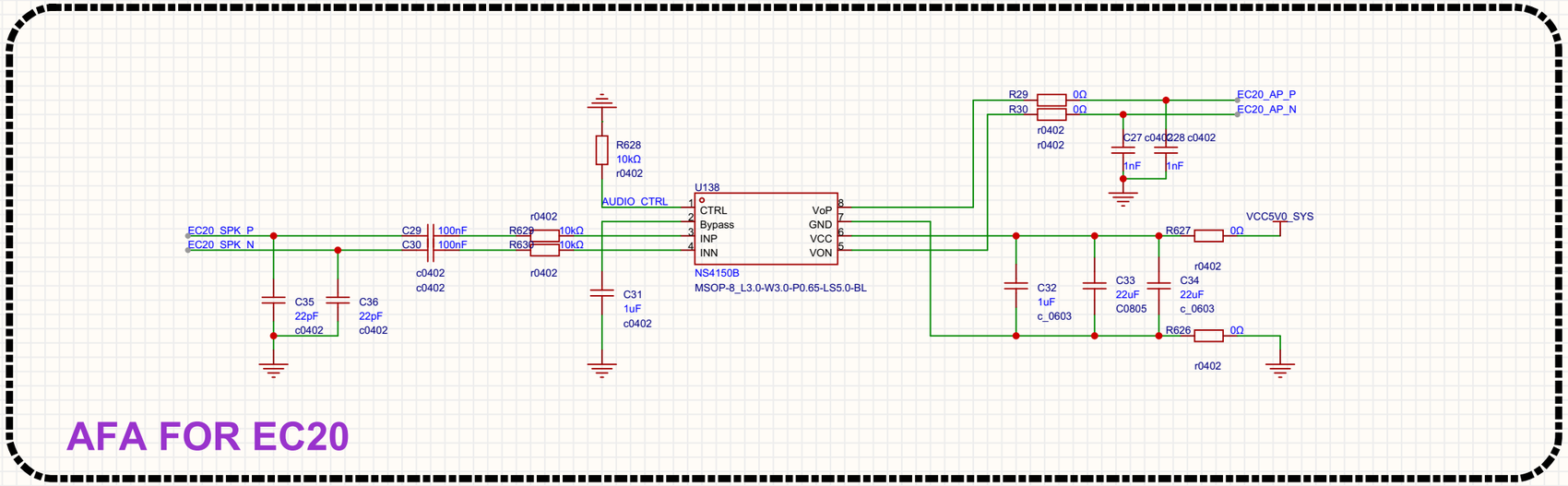


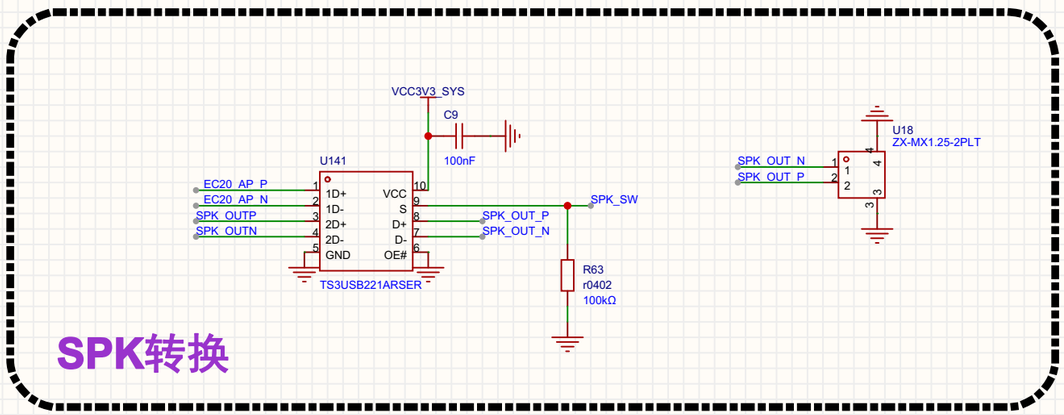
OPTION	WIFI				BT	Crystals	VCCIO_SDIO	OPTION1	OPTION2	OPTION3	OPTION4	OPTION5
	a	b/g/n	ac	5GHz								
AW-CM256SM	Yes	Yes	Yes	Yes	4.2	37.4MHz	1.71-3.6V	Yes	Yes	Yes@SDIO2.0 No@SDIO3.0	No	No
AP6236/AP6212	No	Yes	No	No	4.2/4.0	26MHz	1.71-3.6V	Yes	Yes	No	No	No
AP6256/AP6255	Yes	Yes	Yes	Yes	5.0/4.2	37.4MHz	1.71-3.6V	Yes	Yes	Yes@SDIO2.0 No@SDIO3.0	No	No
RTL8189ETV Module F89FTSM12-W3	No	Yes	No	No	No	Module Integrated	1.8-3.3V	No	No	No	No	No
RTL8723DS Module 6223A-SRD	No	Yes	No	No	4.2	Module Integrated	1.62-3.6V	No	No	No	Yes	Yes
RTL8821CS Module 6221A-SRC	Yes	Yes	Yes	Yes	4.2	Module Integrated	1.7-3.45V	No	No	No	No	No

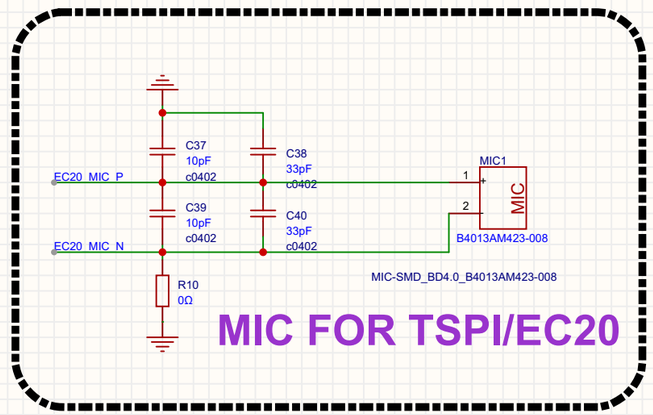

LPDDR4_DQ0_A
LPDDR4_DQ1_A
LPDDR4_DQ2_A
LPDDR4_DQ3_A
LPDDR4_DQ4_A
LPDDR4_DQ5_A
LPDDR4_DQ6_A
LPDDR4_DQ7_A
LPDDR4_DQS0P_A
LPDDR4_DQS0N_A
LPDDR4_DMI0_A
LPDDR4_DQ8_A
LPDDR4_DQ9_A
LPDDR4_DQ10_A
LPDDR4_DQ11_A
LPDDR4_DQ12_A
LPDDR4_DQ13_A
LPDDR4_DQ14_A
LPDDR4_DQ15_A
LPDDR4_DQS1P_A
LPDDR4_DQS1N_A
LPDDR4_DMI1_A
LPDDR4_A0_A
LPDDR4_A1_A
LPDDR4_A2_A
LPDDR4_A3_A
LPDDR4_A4_A
LPDDR4_A5_A
LPDDR4_CLKP_A
LPDDR4_CLKN_A
LPDDR4_CKE0_A
LPDDR4_CKE1_A
LPDDR4_CSN0_A
LPDDR4_CSN1_A
LPDDR4_RESET
LPDDR4_ODT0_CA_A
LPDDR4_ODT0_CA_B
LPDDR4_DQ0_B
LPDDR4_DQ1_B
LPDDR4_DQ2_B
LPDDR4_DQ3_B
LPDDR4_DQ4_B
LPDDR4_DQ5_B
LPDDR4_DQ6_B
LPDDR4_DQ7_B
LPDDR4_DQS0P_B
LPDDR4_DQS0N_B
LPDDR4_DMI0_B
LPDDR4_DQ8_B
LPDDR4_DQ9_B
LPDDR4_DQ10_B
LPDDR4_DQ11_B
LPDDR4_DQ12_B
LPDDR4_DQ13_B
LPDDR4_DQ14_B
LPDDR4_DQ15_B
LPDDR4_DQS1P_B
LPDDR4_DQS1N_B
LPDDR4_DMI1_B
LPDDR4_A0_B
LPDDR4_A1_B
LPDDR4_A2_B
LPDDR4_A3_B
LPDDR4_A4_B
LPDDR4_A5_B
LPDDR4_CLKP_B
LPDDR4_CLKN_B
LPDDR4_CKE0_B
LPDDR4_CKE1_B
LPDDR4_CSN0_B
LPDDR4_CSN1_B
U114.1
CH A
CH B
DQ0_A
DQ1_A
DQ2_A
DQ3_A
DQ4_A
DQ5_A
DQ6_A
DQ7_A
DQS0_T_A
DQS0_C_A
DMI0_A
DQ8_A
DQ9_A
DQ10_A
DQ11_A
DQ12_A
DQ13_A
DQ14_A
DQ15_A
DQS1_T_A
DQS1_C_A
DMI1_A
CA0_A
CA1_A
CA2_A
CA3_A
CA4_A
CA5_A
CK_T_A
CK_C_A
CKE0_A
CKE1_A
CKE2_A_NC
CS0_A
CS1_A
CS2_A_NC
ODT_CA_A
ZQ0
ZQ1
ZQ2_NC
DQ0_B
DQ1_B
DQ2_B
DQ3_B
DQ4_B
DQ5_B
DQ6_B
DQ7_B
DQS0_T_B
DQS0_C_B
DMI0_B
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DQ9_B
DQ10_B
DQ11_B
DQ12_B
DQ13_B
DQ14_B
DQ15_B
DQS1_T_B
DQS1_C_B
DMI1_B
CA0_B
CA1_B
CA2_B
CA3_B
CA4_B
CA5_B
CK_T_B
CK_C_B
CKE0_B
CKE1_B
CKE2_B_NC
CS0_B
CS1_B
CS2_B_NC
ODT_CA_B
RESET_n
VDDQ_DRAM
R605
R604
240Ω
C563
1nF
U114.2
VDD1_1V8_DDR
VDD2_DDR
VDDQ_DRAM
VDD1
1.8V
VDD2
1.1V
VDDQ
LP4
1.1V
LP4x
0.6V
VDD1_1
VDD1_2
VDD1_3
VDD1_4
VDD1_5
VDD1_6
VDD1_7
VDD1_8
VDD2_1
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DNU_10
DNU_11
DNU_12
LPDDR4X/4
VCCA1V8_PMU
VDD1_1V8_DDR
R402
0Ω
C564
10uF
VDDQ_DRAM
RP1
R612
0Ω
VCC0V6_DDR
0.6V
VDD2_DDR
RP2
R611
0Ω/NC
1.1V
RP1
RP2
LPDDR4
NC
0R
LPDDR4X
0R
NC
default
C868
22uF
C565 C566
10uF 10uF
C567 C568
1uF 1uF
C569 C570 C571 C572 C573 C574 C575 C576 C577 C578 C579 C580
100nF100nF100nF100nF100nF100nF100nF100nF100nF100nF100nF100nF
C869
22uF
C584 C585
10uF 10uF
C586 C587
1uF 1uF
C588 C589 C590 C591 C592 C593 C594 C595 C596 C597 C938
100nF100nF100nF100nF100nF100nF100nF100nF100nF100nF100nF
C598
10uF
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电源管理

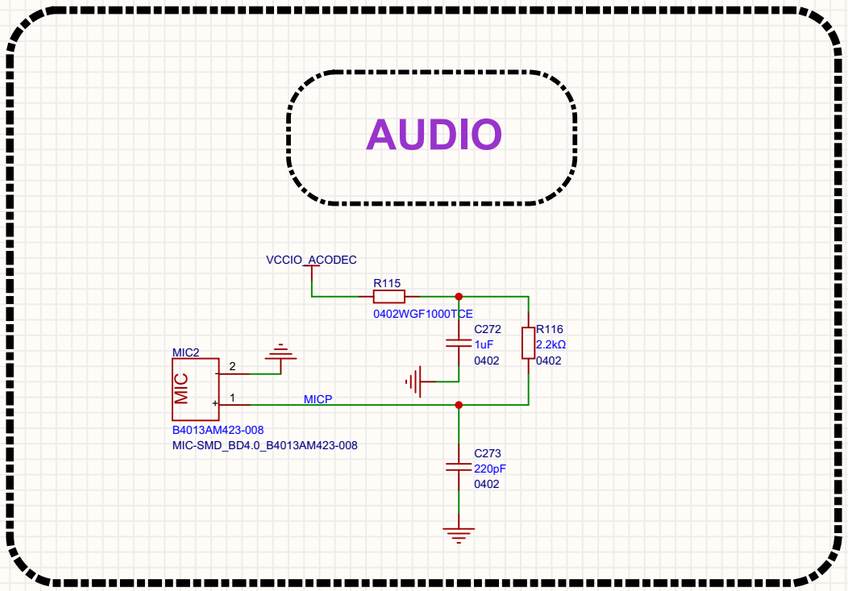

EMMC
VCCIO_FLASH
R424
10kΩ
R425
10kΩ/NC
U120.1
EMMC_D0
EMMC_D1
EMMC_D2
EMMC_D3
EMMC_D4
EMMC_D5
EMMC_D6
EMMC_D7
EMMC_CMD
EMMC_CLKOUT
EMMC_DATA_STROBE
EMMC_RSTN
A3
A4
A5
B2
B3
B4
B5
B6
M5
M6
H5
K5
C2
DATA0
DATA1
DATA2
DATA3
DATA4
DATA5
DATA6
DATA7
CMD
CLK
Data Strobe
RST_n
VDDi
C645
10pF-NC
R426
0Ω
R427
47kΩ/NC
R428
10kΩ
C870
100nF
C646
2.2uF
依据emmc规范
VCCQ1
VCCQ2
VCCQ3
VCCQ4
VCCQ5
C6
M4
N4
P3
P5
C871
100nF
C872
100nF
C649
100nF
C650
100nF
C651
4.7uF
VCC_3V3
VCC1
VCC2
VCC3
VCC4
E6
F5
J10
K9
C647
100nF
C648
100nF
C652
4.7uF
VSS1
VSS2
VSS3
VSS4
VSS5
VSS6
J5
A6
C4
E7
G5
H10
VSSQ1
VSSQ2
VSSQ3
VSSQ4
VSSQ5
K8
N2
N5
P4
P6
VSF4
VSF3
VSF2
VSF1
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F10
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E9
U120.2
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A8
A9
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NC100
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RFU8
RFU7
RFU6
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J12
J3
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J1
H14
H13
H12
H3
H2
H1
P10
P7
K7
K6


UART2_TX_M0_DEBUG
R287
0Ω
DBG_RXD
VCC_UART
UART2_RX_M0_DEBUG
DBG_RXD
DBG_TXD
H3
1
2
3
GND
MH127V-11-03-H22-0820
R18
10kΩ
UART2_RX_M0_DEBUG
BSN20-7
Q3
DBG_TXD
VCC_UART
R23
0Ω
VCC_3V3
信号防干扰电路

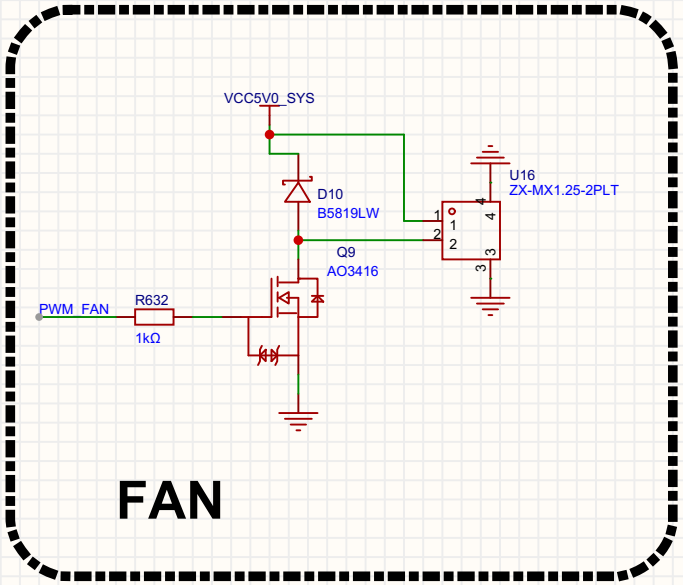

R628
10kΩ
r0402
AUDIO_CTRL
U138
CTRL
Bypass
INP
INN
VoP
GND
VCC
VON
NS4150B
MSOP-8_L3.0-W3.0-P0.65-LS5.0-BL
EC20_SPK_P
EC20_SPK_N
C29 100nF
C30 100nF
c0402
R629 10kΩ
R630 10kΩ
r0402
C35 22pF c0402
C36 22pF c0402
C31 1uF c0402
R29 0Ω
R30 0Ω
r0402
EC20_AP_P
EC20_AP_N
C27 c0402 1nF
C28 c0402 1nF
C32 1uF c_0603
C33 22uF C0805
C34 22uF c_0603
R627 0Ω
R626 0Ω
VCC5V0_SYS
AFA FOR EC20

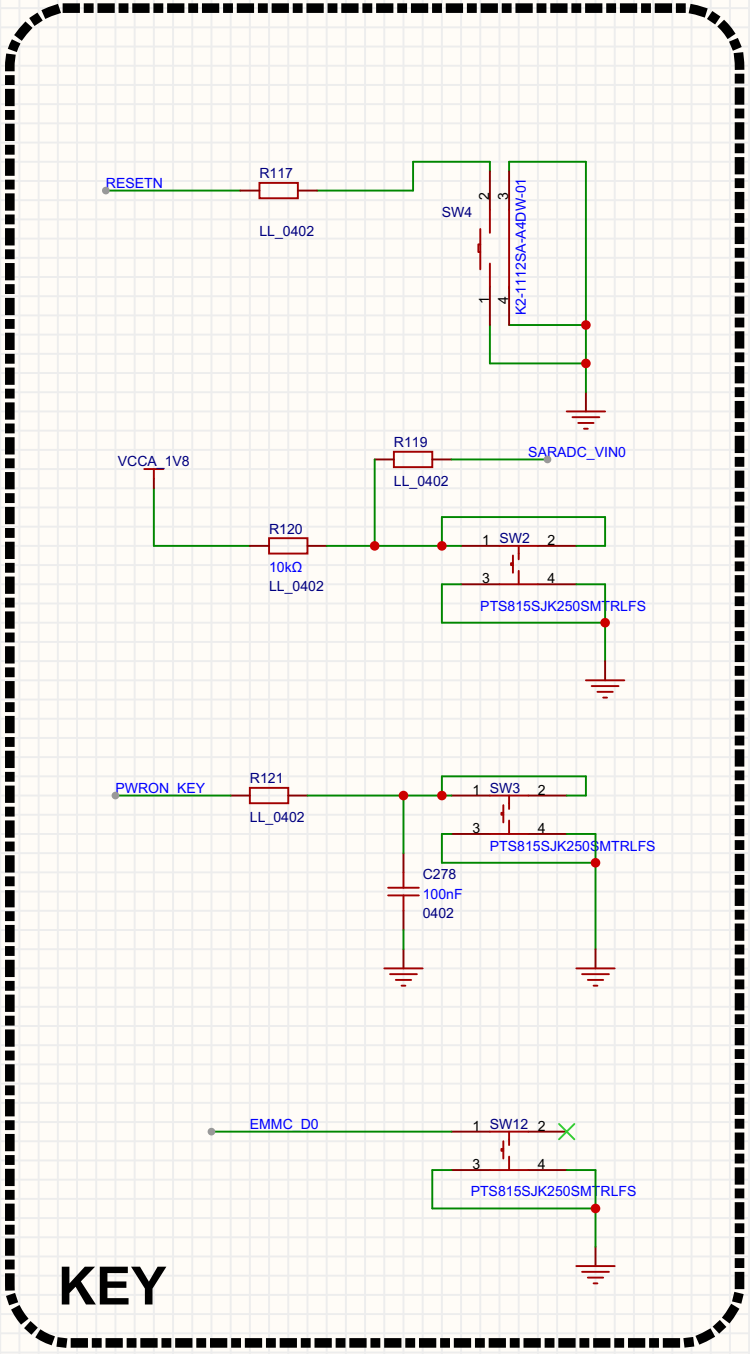

VCC3V3_SYS
C9
100nF
U141
1D+
1D-
2D+
2D-
GND
VCC
S
D+
D-
OE#
TS3USB221ARSER
EC20_AP_P
EC20_AP_N
SPK_OUTP
SPK_OUTN
SPK_SW
SPK_OUT_P
SPK_OUT_N
U18
ZX-MX1.25-2PLT
R63
r0402
100kΩ
SPK转换

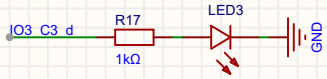

C37 10pF c0402
C38 33pF c0402
C39 10pF c0402
C40 33pF c0402
EC20_MIC_P
EC20_MIC_N
MIC1
MIC
B4013AM423-008
MIC-SMD_BD4.0_B4013AM423-008
R10
0Ω
MIC FOR TSPI/EC20


AUDIO
VCCIO_ACODEC
R115
0402WGF1000TCE
C272 1uF 0402
R116 2.2kΩ 0402
MIC2
MIC
MICP
B4013AM423-008
MIC-SMD_BD4.0_B4013AM423-008
C273 220pF 0402

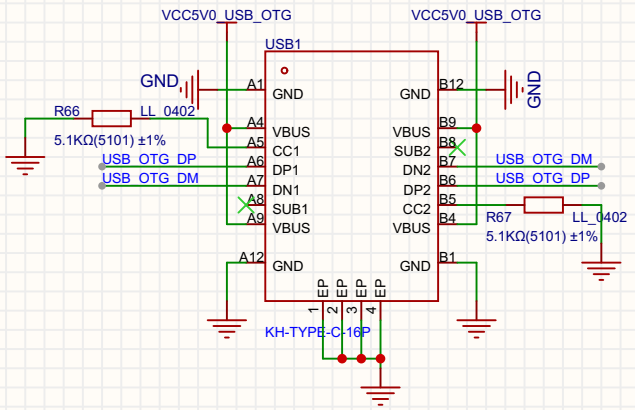
U126 U127 U128 U129



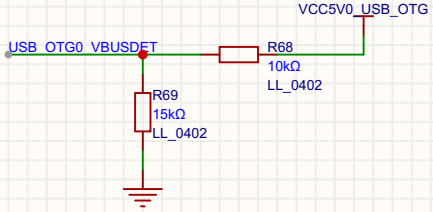




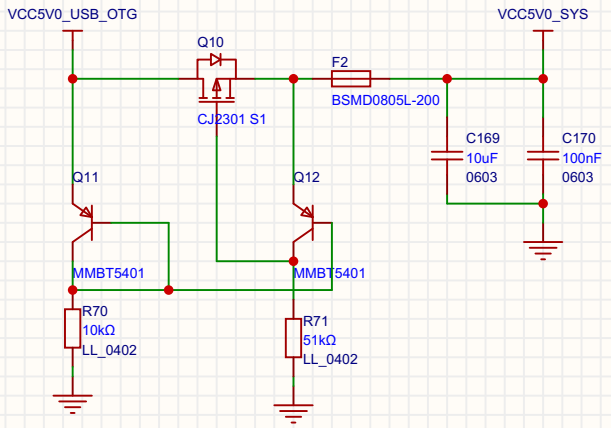
5V POWER IN

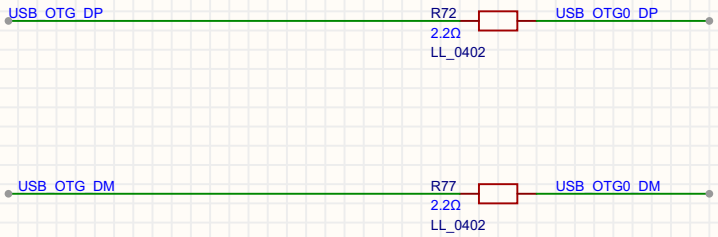


OTG_VBUSDET

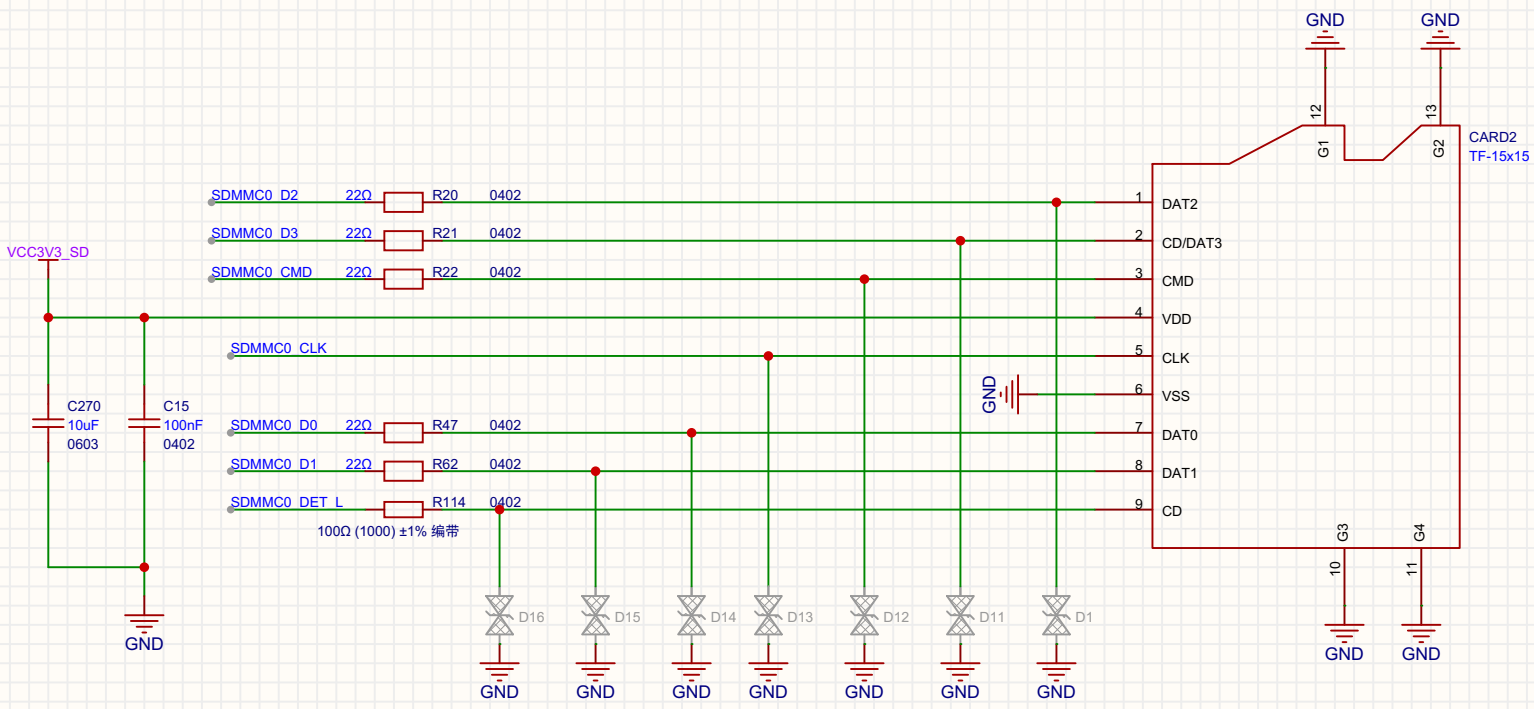


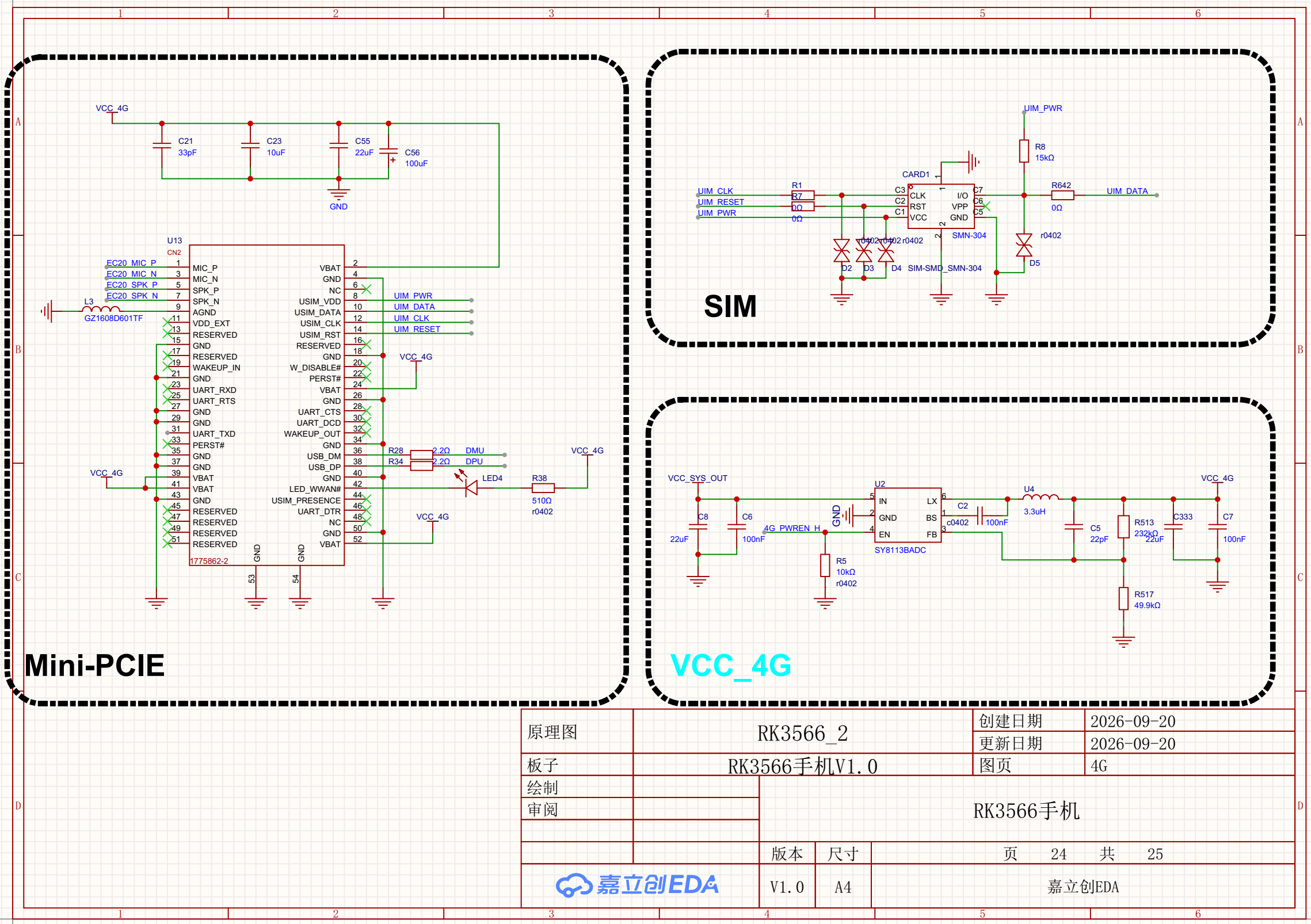
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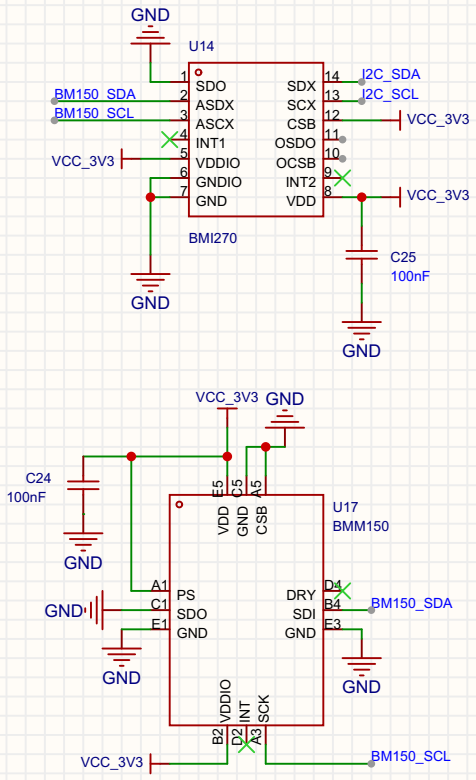


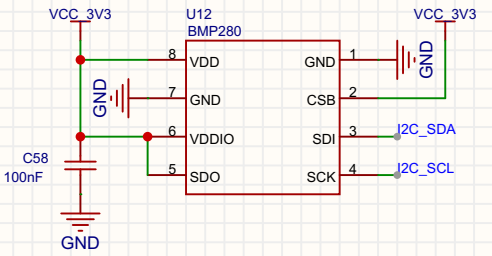
MicroSD Card





原理图	RK3566_2	创建日期	2026-09-20
		更新日期	2026-09-20
板子	RK3566手机V1.0	图页	4G
绘制			
审阅		RK3566手机	
		版本	尺寸	页 24 共 25
嘉立创EDA		V1.0	A4	嘉立创EDA





9 Axis

原理图	RK3566_2	创建日期	2026-09-23
		更新日期	2026-09-23
板子	RK3566手机V1.0	图页	Sensor
绘制		RK3566手机	
审阅			
		版本	尺寸	页 25 共 25
嘉立创EDA		V1.0	A4	嘉立创EDA